

23

Chemoselectivity and protecting groups

Connections

➡ Building on

- Carbonyl addition and substitution [ch6](#), [ch10](#), & [ch12](#)
- Oxidation of alcohols [ch9](#)
- Mechanisms and catalysis [ch12](#)
- Electrophilic addition to alkenes [ch19](#)

Arriving at

- Regio-, stereo-, and chemoselectivity
- Reagents for reduction of alkenes and carbonyl compounds
- Removal of functional groups
- Reduction of benzene rings
- Reagents for oxidation of alcohols
- Reagents for oxidation of alkenes
- Protection of aldehydes, ketones, alcohols, and amines
- Synthesis of peptides

➡ Looking forward to

- Regioselectivity [ch24](#)
- Reactions of enolates [ch25](#) & [ch26](#)
- Sulfur chemistry [ch27](#)
- Retrosynthetic analysis [ch28](#)
- Cycloadditions [ch34](#)

Selectivity

Most organic molecules contain more than one functional group, and most functional groups can react in more than one way, so organic chemists often have to predict *which* functional group will react, *where* it will react, and *how* it will react. These questions are what we call *selectivity*.

Selectivity comes in three sorts: chemoselectivity, regioselectivity, and stereoselectivity. Chemoselectivity is *which* group reacts; regioselectivity is *where* it reacts. Stereoselectivity is *how* the group reacts with regard to the stereochemistry of the product.

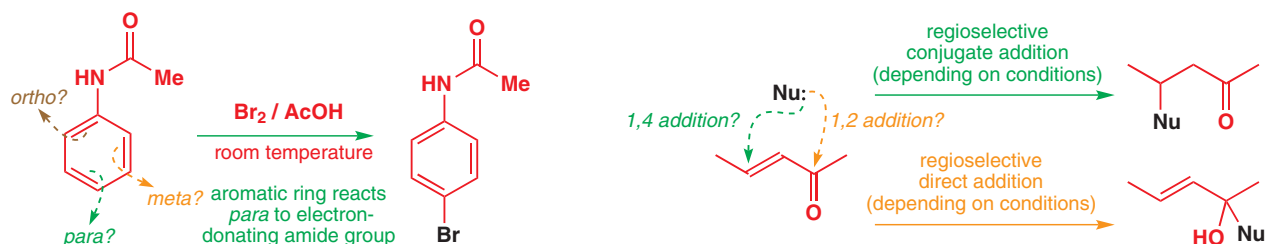
● Selectivity

There are three main types of selectivity:

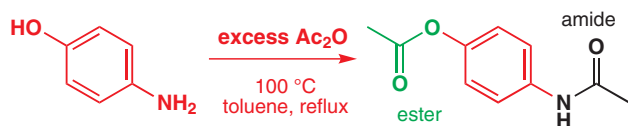
- chemoselectivity: *which* functional group will react (this chapter)
- regioselectivity: *where* it will react (Chapter 24)
- stereoselectivity: *how* it will react (stereochemistry of the products) (Chapters 32, 33, and 41)

We have talked a lot about regioselectivity, without calling it that, in the last two chapters. In Chapter 21 you learned how to predict and explain which product(s) you get from electrophilic aromatic substitution reactions. The functional group is the aromatic ring: *where* it reacts is the reaction's regioselectivity. In Chapter 22 you saw that nucleophilic addition to an unsaturated ketone can take place in a 1,2- or 1,4-fashion—the question of which happens (*where* the unsaturated ketone reacts) is a question of regioselectivity. We will address regioselectivity in much more detail in the next chapter.

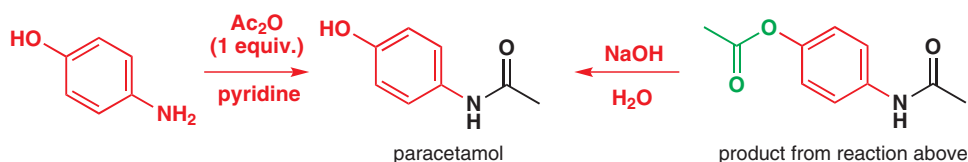
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.



But this chapter is about *chemoselectivity*—in a compound with more than one functional group, which group reacts? Let's start with a straightforward example—the synthesis of the painkiller paracetamol. 4-Aminophenol could react with acetic anhydride at both nitrogen and oxygen to give a compound containing an amide and an ester functional group. This is what happens on heating with excess acetic anhydride (Ac_2O) in toluene.



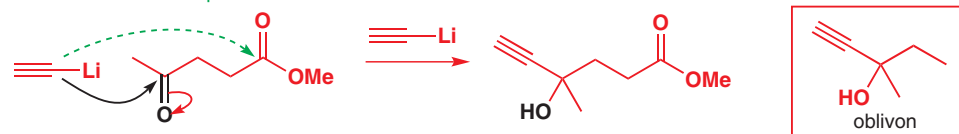
But with just one equivalent of acetic anhydride in the presence of a base (pyridine) only the NH_2 group is acylated, and paracetamol is the product. This is chemoselectivity, and it is to be expected that the NH_2 group is more nucleophilic than the OH group. It is even possible to hydrolyse the doubly acetylated product to paracetamol with aqueous sodium hydroxide. The ester is more reactive than the amide and hydrolyses much more easily. This is another chemoselective reaction.



➡ Why amines are more nucleophilic than alcohols and why esters are more reactive than amides was all explained in Chapter 10.

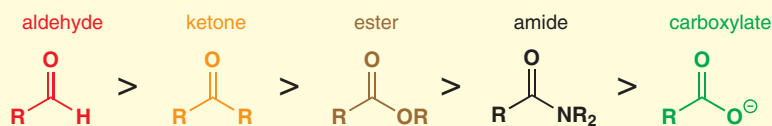
We know that ketones are more reactive towards Grignard reagents and organolithiums than esters because you can't isolate a ketone from the reaction of an ester with a Grignard reagent or an organolithium. Chemists at the pharmaceutical company Pfizer made use of this fact while they were developing anticonvulsants related to the tranquillizer oblivon. By adding lithium acetylide to ketones, they were able to make a tertiary alcohol by chemoselective reaction of a ketone in the presence of an ester.

ketone is more electrophilic than ester



These last two reactions work because, although each starting material contains two carbonyl groups, one is more electrophilic and therefore more reactive towards nucleophiles (OH^- in the first case; lithium acetylide in the second) than the other. We can order carbonyl compounds into a sequence in which it will *usually* be possible to react those on the left with nucleophiles in the presence of those on the right.

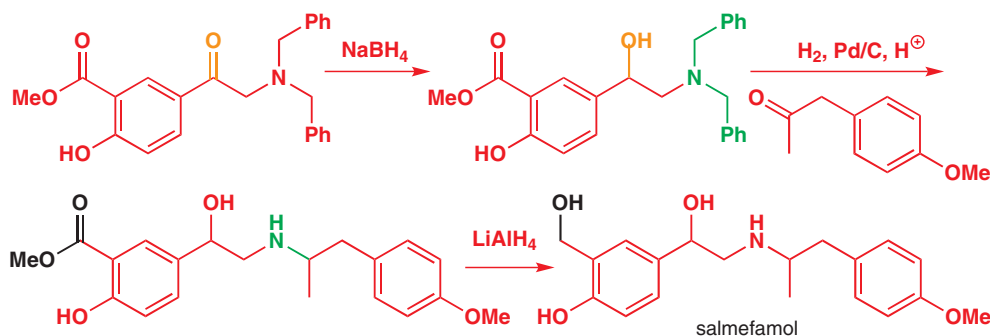
• Reactivity towards nucleophiles



We've already discussed this sequence of reactivity in relation to acid derivatives in Chapter 10—make sure you understand the reason for the ordering of ester > amide > carboxylate. Here we're adding to the list aldehyde (the most reactive, for steric reasons—it is the least hindered) and ketone (more reactive than esters because the carbonyl group is not stabilized by conjugation with a lone pair).

Reducing agents

Chemists at Glaxo exploited this reactivity sequence in their synthesis of the anti-asthma drug salmefamol (sister compound to the bestseller salbutamol). Three reducing agents are used in the sequence: sodium borohydride (NaBH_4), hydrogen gas over a palladium catalyst, and lithium aluminium hydride (LiAlH_4).



Why not use LiAlH_4 all the time?

In general, it's best to use the mildest conditions possible for any particular reaction—the potential for unwanted side-reactions is lessened. What is more, NaBH_4 is a lot easier to handle than LiAlH_4 —for example, it simply dissolves in water while LiAlH_4 catches fire if it gets wet. NaBH_4 is usually used to reduce aldehydes and ketones, even though LiAlH_4 also works.

We shall use this synthesis as a basis for discussion on chemoselectivity in reductions. In the first step, sodium borohydride leaves the carbonyl group of the ester untouched while it reduces the ketone (in orange); in the last step, lithium aluminium hydride reduces the ester (in black). These chemoselectivities are typical of these two most commonly used reducing agents: borohydride can usually be relied upon to reduce an aldehyde or a ketone in the presence of an ester, while lithium aluminium hydride will reduce almost any carbonyl group.

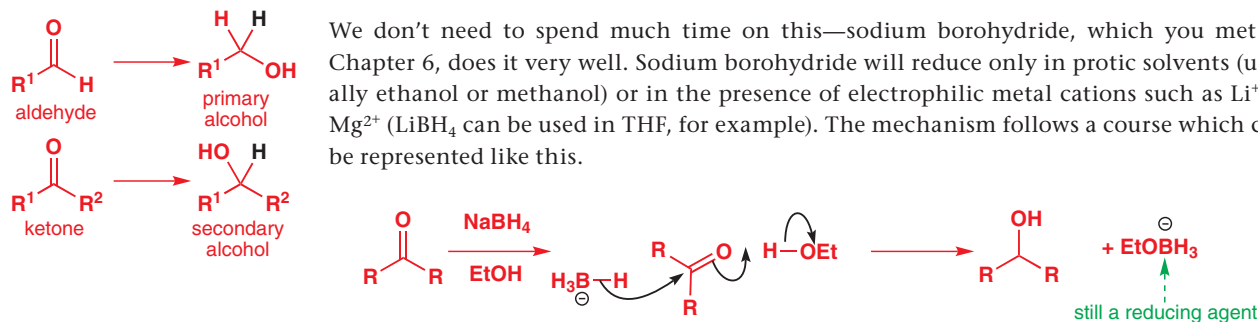
Reduction of carbonyl groups

■ It is worth noticing that *all* of the reducing agents for carbonyl groups described in this section are hydrides of boron or aluminium.

We should now look in detail at reductions of carbonyl compounds, and in doing so we shall introduce a few more specialized reducing agents. Then we will come back to the other type of reduction in the salmefamol synthesis—catalytic hydrogenation.

How to reduce aldehydes and ketones to alcohols

We don't need to spend much time on this—sodium borohydride, which you met in Chapter 6, does it very well. Sodium borohydride will reduce only in protic solvents (usually ethanol or methanol) or in the presence of electrophilic metal cations such as Li^+ or Mg^{2+} (LiBH_4 can be used in THF, for example). The mechanism follows a course which can be represented like this.

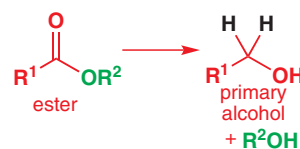
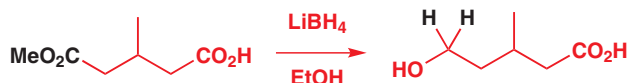


The essence of the reaction is the transfer of a hydrogen atom with two electrons (called **hydride transfer**, although no hydride ion is involved) from boron to carbon. The developing

negative charge on oxygen is protonated by the alcohol, and resulting alkoxide adds to the boron during or immediately after the reduction. The by-product, an alkoxyborohydride anion, is itself a reducing agent, and can go on to reduce three more molecules of carbonyl compound, transferring step-by-step all of its hydrogen atoms.

How to reduce esters to alcohols

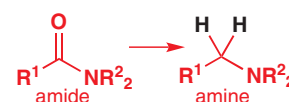
LiAlH_4 is often the best reagent, and gives alcohols by the mechanism we discussed in Chapter 10 (p. 217). As a milder alternative (needed because LiAlH_4 has caused countless fires through careless handling), lithium borohydride in alcoholic solution will reduce esters—in fact, it has useful selectivity for esters over acids or amides that LiAlH_4 does not have. *Sodium* borohydride reduces most esters only very slowly.



■ Why not try writing the mechanism out now for this reaction to make sure you understand it, before checking back to p. 217. In a moment, we will show you a slightly more sophisticated version, in which we account for the fate of the Li and Al species.

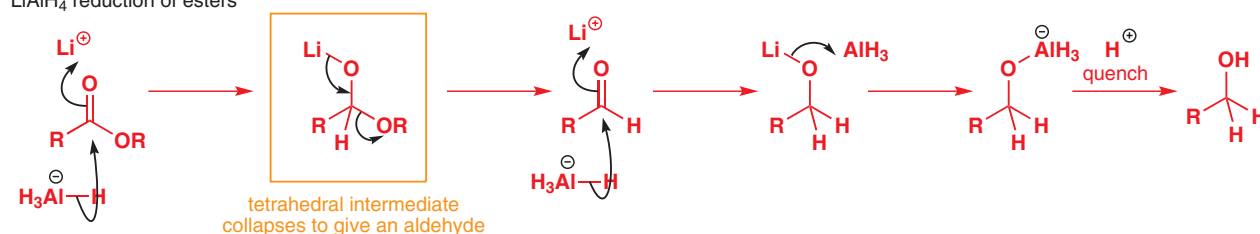
How to reduce amides to amines

Again, LiAlH_4 is a good reagent for this transformation. The mechanism follows a similar course to the reduction of esters: both are written out in detail below, and there is a key difference at the steps boxed in orange and in green. In the orange box, loss of alkoxide from the tetrahedral intermediate forms an aldehyde, which is reduced further. This doesn't happen with amides; instead the anionic oxygen is lost—assisted by coordination to aluminium—to form an iminium ion. A good alternative for the reduction of amides to amines is borane (BH_3), described in the next section.

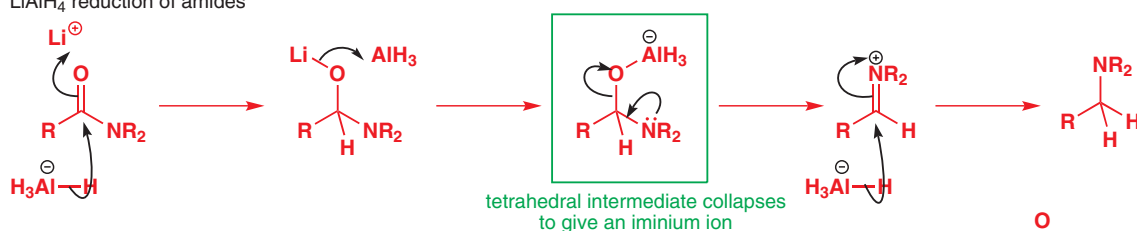


■ The mechanism for ester reduction here has rather more detail than the simplified one we presented to you in Chapter 10.

LiAlH_4 reduction of esters



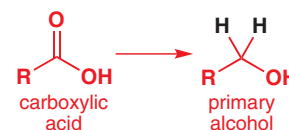
LiAlH_4 reduction of amides



How to reduce carboxylic acids to alcohols

The best reagent for this is borane, BH_3 . Borane is, in fact, a gas with the structure B_2H_6 , but it can be 'tamed' as a liquid by complexing it with ether (Et_2O), THF, or dimethyl sulfide (DMS , Me_2S).

Although borane appears superficially similar to borohydride, it is not charged, and that makes all the difference to its reactivity. Whereas borohydride reacts best with the most electrophilic carbonyl groups, borane's reactivity is dominated by its desire to accept an electron pair into the boron's empty p orbital. In the context of carbonyl group reductions, this means that **borane reduces electron-rich carbonyl groups fastest**. The carbonyl groups of

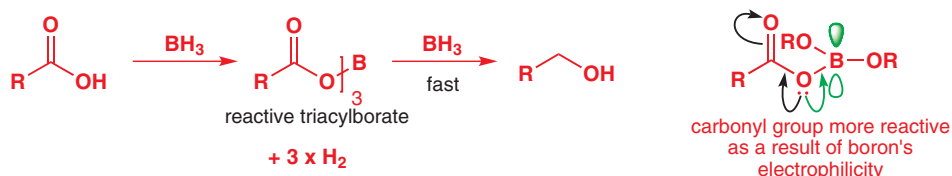


These complexes are formed when BH_3 , a Lewis acid, accepts a lone pair of electrons from the ether or sulfide—a Lewis base. Lewis acids and bases were described on p. 180.

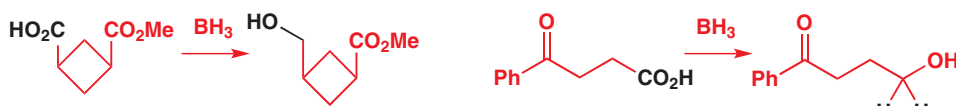
■ This 'sharing' of lone pair between two conjugating groups is also the reason why anhydrides are more reactive than esters (see p. 206).

acyl chlorides and esters are relatively electron-poor (Cl and OR are very electronegative); borane will not touch acyl chlorides and reduces esters only slowly. But it will reduce very effectively both carboxylic acids and amides.

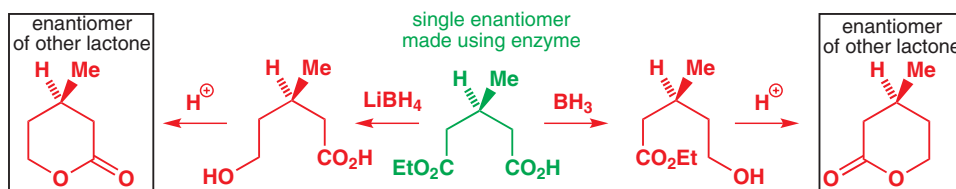
Borane reacts with carboxylic acids first of all to form triacylborates, with evolution of hydrogen gas. Esters are usually less electrophilic than ketones because of conjugation between the carbonyl group and the lone pair of the sp^3 hybridized oxygen atom—but, in these boron esters, the oxygen next to the boron has to share its lone pair between the carbonyl group and the boron's empty p orbital, so they are considerably more reactive than normal esters.



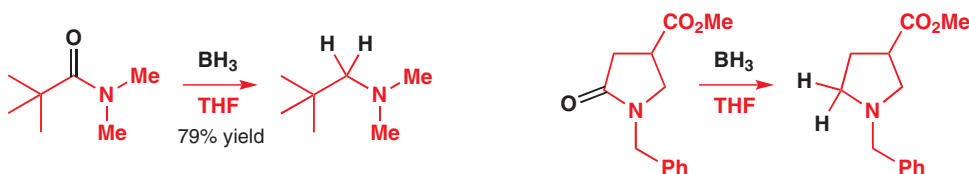
Borane is a highly chemoselective reagent for the reduction of carboxylic acids in the presence of other reducible functional groups such as esters, and even ketones.



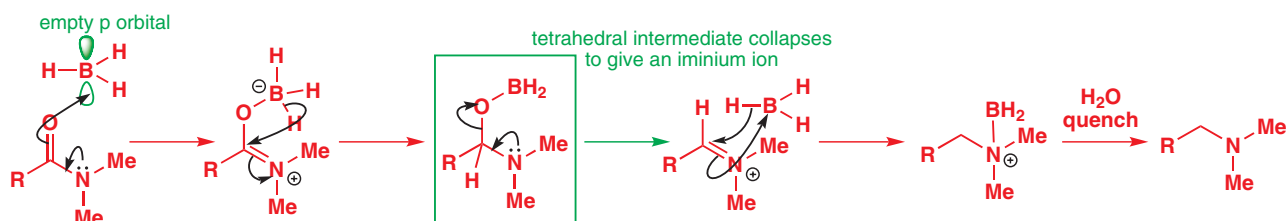
Borane and lithium borohydride are a most useful pair of reducing agents, with opposite selectivities. Japanese chemists used an enzyme to make a single enantiomer of the acid below, and were able to reduce either the ester or the carboxylic acid by choosing lithium borohydride or borane as their reagent. Check for yourself that the lactones (cyclic esters) in black frames are enantiomers.



Because borane reacts well with electron-rich carbonyl groups, it is also a good alternative to LiAlH_4 for reducing amides to amines, and is chemoselective even in the presence of an ester:

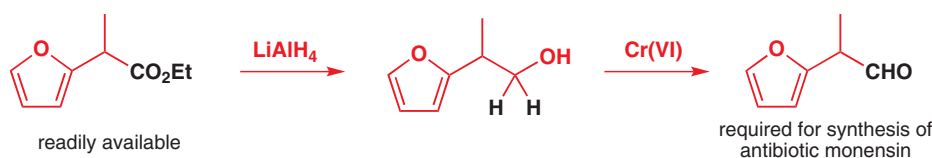
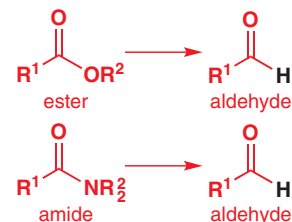


The carbonyl group of an amide is electron-rich because it receives electron density from the delocalized N lone pair. It therefore complexes well with the empty p orbital of the Lewis acidic borane. Hydride transfer is then possible from anionic boron to electrophilic carbon. The resulting tetrahedral intermediate collapses to an iminium ion that is reduced again by the borane.



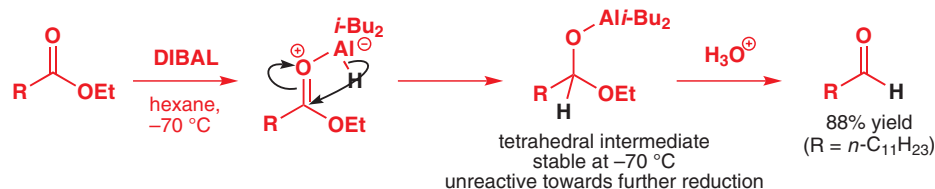
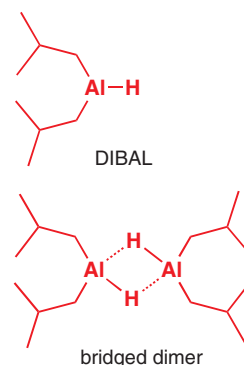
How to reduce esters or amides to aldehydes

The step boxed in orange in the ester reduction scheme on p. 531 gave an aldehyde. The aldehyde is more readily reduced than the ester, so the reduction doesn't stop there, but carries on to the alcohol oxidation level. How, then, can you reduce an ester to an aldehyde? This is a real problem in synthetic chemistry—the ester below, for example, is easy to make by methods you will meet in Chapter 25, but an important synthesis of the antibiotic monensin requires the aldehyde.



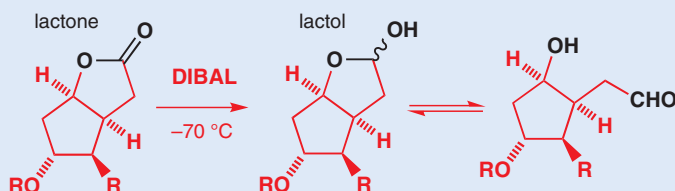
In this case, the chemists decided simply to put up with the fact that LiAlH_4 gives the alcohol, and oxidize the alcohol back to the aldehyde using chromium(VI), the oxidant you met in Chapter 9 (p. 194). There is, however, a reagent that will sometimes do the job in a single step, although you must bear in mind that this is not at all a general reaction. The reagent is known as DIBAL or DIBALH—diisobutyl aluminium hydride, $i\text{-Bu}_2\text{AlH}$.

DIBAL is an alane: its structure is shown in the margin. Its chemistry is in many ways like borane—it exists as a bridged dimer, and it becomes a reducing agent only after it has formed a Lewis acid–base complex, so like borane it too reduces electron-rich carbonyl groups most rapidly. DIBAL will reduce esters even at -70°C , and at this temperature the tetrahedral intermediate, formed by the transfer of hydride from aluminium to carbon (and shown below), may be stable. Only in the aqueous work-up does it collapse to the aldehyde. This step also destroys any excess DIBAL so no further reduction is possible.



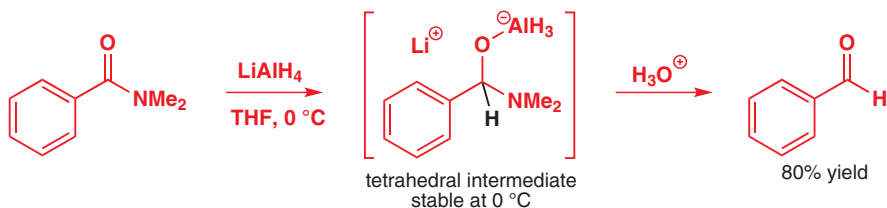
Lactols from lactones

A stable tetrahedral intermediate is more likely in the reduction of lactones, for the same reasons that cyclic hemiacetals are more stable than acyclic ones. DIBAL is most reliable in the reduction of lactones to cyclic hemiacetals (also known as lactols), as in this reaction from E. J. Corey's synthesis of the prostaglandins.



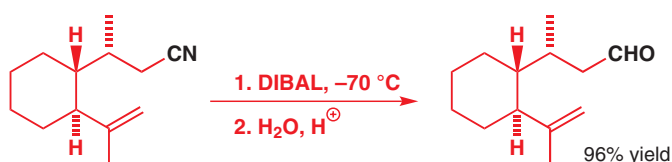
Higher temperatures are needed to promote collapse to the iminium.

In the amide reduction scheme on p. 533, the step framed in green gives an iminium ion. Stopping the reaction before the iminium ion forms would therefore provide a way of making aldehydes from amides because in the absence of the aluminium, this tetrahedral intermediate collapses to an aldehyde. Because tetrahedral intermediates like these formed during amide reduction are rather more stable than those from ester reduction, this can often be achieved simply by carrying out the amide reduction and quenching with water, all at 0 °C.



DIBAL is also good for reducing nitriles to aldehydes. Indeed, this reaction and the reduction of lactones to lactols (see box above) are the best things that DIBAL does.

Carboxylic acids can be reduced to aldehydes via their acyl chlorides using the Rosenmund reaction—see below.



The box below summarizes the chemoselectivity of all of these reagents.

Also included is sodium cyanoborohydride, which as you saw in Chapter 11 reduces imines but not carbonyl compounds.

Summary of reducing agents for carbonyl groups

Key

reduced

reduced slowly

not usually reduced

$\text{R}-\text{C}(=\text{O})-\text{H}$

via alcohol or DIBAL

$\text{R}-\text{C}(=\text{O})-\text{H}$

LiAlH₄, 0 °C

$\text{R}-\text{C}(=\text{O})-\text{H}$

via acyl chloride (see later)

← most reactive
least reactive →

	$\text{R}-\text{C}(=\text{NHR})-\text{H}$ imine	$\text{R}-\text{C}(=\text{O})-\text{H}$ aldehyde	$\text{R}-\text{C}(=\text{O})-\text{R}$ ketone	$\text{R}-\text{C}(=\text{O})-\text{OR}$ ester	$\text{R}-\text{C}(=\text{O})-\text{NR}_2$ amide	$\text{R}-\text{C}(=\text{O})-\text{OH}$ carboxylic acid
NaCNBH₃	Green	Orange	Orange	Red	Red	Red
NaBH₄	Green	Green	Green	Orange	Red	Red
LiBH₄	Green	Green	Green	Green	Red	Red
LiAlH₄	Green	Green	Green	Green	Green	Orange
BH₃	Green	Orange	Orange	Orange	Green	Green

$\text{R}-\text{CH}_2-\text{NHR}$

$\text{R}-\text{CH}_2-\text{OH}$

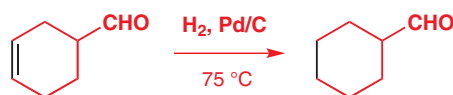
$\text{R}-\text{CH}(\text{OH})-\text{R}$

$\text{R}-\text{CH}_2-\text{OH}$

$\text{R}-\text{CH}_2-\text{NR}_2$

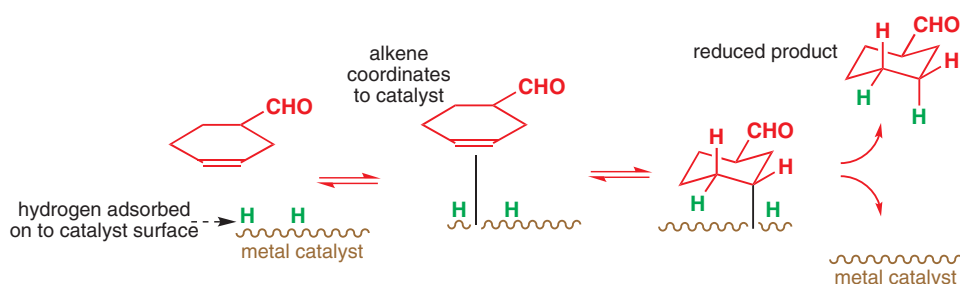
$\text{R}-\text{CH}_2-\text{OH}$

a reducing agent for other, weaker, double and triple bonds, such as $C=C$, $C=N$, $C\equiv C$ and $C\equiv N$. To do these reactions, a metal catalyst is required and the process is known as catalytic hydrogenation. The hydrogen is provided by a cylinder, perhaps via a balloon, or can be made by electrolysis and pumped with the substrate over the catalyst. In the example below, the alkene is reduced while the aldehyde remains untouched.



The catalysts used to make hydrogen react with double bonds are transition metals: usually palladium (as in this example) or platinum, but sometimes nickel, rhodium, or ruthenium. We will talk about several different reductions in this section, but the mechanisms of all of them are similar, and very different from those involved in carbonyl reductions.

Catalytic hydrogenation takes place on the surface of the metal. The metal must therefore be finely divided, and is usually dispersed on the surface of an inert support. This is what 'Pd/C' means—finely divided palladium carried on a charcoal support. The first step is chemical absorption of hydrogen onto the metal surface, a process that results in breakage of the H–H bonds and distributes hydrogen atoms where they can react with the organic substrate. Now the alkene can also bond to the metal, and hydrogen can be transferred from the metal to the alkene.



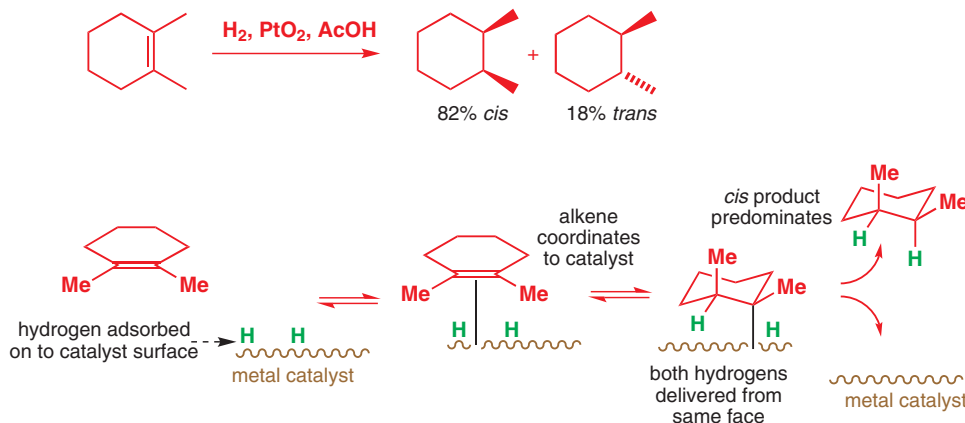
Palladium on charcoal

Palladium on charcoal (Pd/C) is usually 5–10% by mass Pd and 90–95% C. It is made by suspending charcoal powder in a $PdCl_2$ solution, and then reducing the $PdCl_2$ to Pd metal, usually with H_2 gas, but sometimes with formaldehyde, $HCHO$ (which is oxidized to formic acid, HCO_2H). The palladium metal precipitates on to the charcoal, which can be filtered off and dried. The fine Pd particles present maximum surface area to the reaction they catalyse and, while Pd is an expensive metal, it is recyclable since the Pd/C is insoluble and can be recovered by filtration.

➔ We will look in more detail at the ways alkenes bond with metals in Chapter 40.

How to reduce alkenes to alkanes

Hydrogenation with a palladium or a platinum catalyst is the most common way of reducing alkenes. You may find our mechanism rather unsatisfactory, but it is hard to draw curly arrows for the reactions involved here. There is, however, plenty of evidence that the hydrogenation occurs like this, for example the alkene below reacts in such a way that the major product receives both hydrogen atoms on the same face of the molecule—just what we would expect from a reaction at a surface.

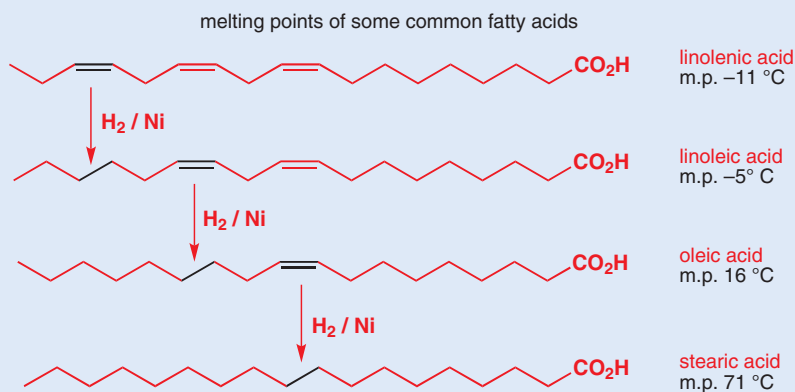


■ PtO_2 , platinum oxide, is known as Adams' catalyst. The actual catalyst is not the oxide but Pt metal formed by reduction of PtO_2 to Pt during the hydrogenation.

Hydrogenated vegetable oil

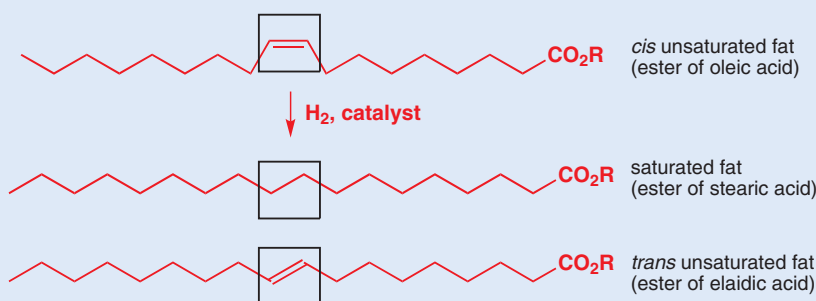
Plants such as soya, rapeseed, cottonseed, and sunflower are useful sources of edible vegetable oils, but these oils are unsuitable as butter substitutes because of their low melting points. Their low melting points relative to animal fats are largely due to *cis* double bonds that disrupt the packing of the alkyl chains in

the solid state. Treating the crude vegetable oil with hydrogen over a metal catalyst removes some of these double bonds, increases the proportion of saturated fat in the oil, and raises its melting point, making it suitable for making margarine.



The reaction is usually stopped before all the double bonds are hydrogenated, of course: margarine manufacturers are desperate to tell us that their products are still 'high in unsaturated fatty acids'. Many also advertise that they are 'low

in *trans* unsaturated fatty acids' because of a suggested link between incidence of coronary heart disease and *trans* unsaturated fatty acid intake.

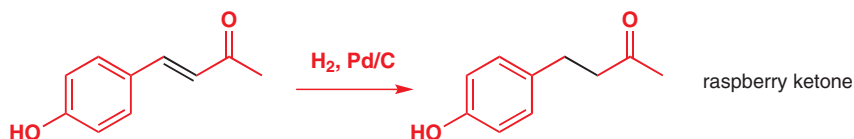


Where have these *trans* double bonds come from? Well, partial hydrogenation can lead to significant double-bond isomerization, not just to regioisomers but

to geometrical isomers too.

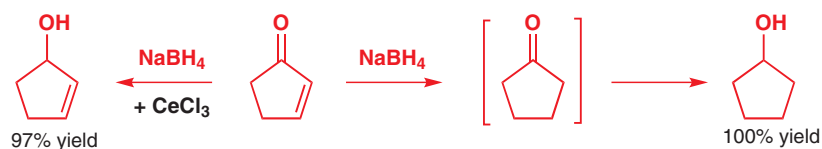
How to reduce α,β -unsaturated carbonyl compounds

It should not now surprise you that regioselective reduction of the $\text{C}=\text{C}$ double bond of an α,β -unsaturated carbonyl compound is best done using catalytic hydrogenation as the $\text{C}=\text{C}$ bond is more susceptible to hydrogenation than the $\text{C}=\text{O}$ bond. The flavouring compound known as raspberry ketone is made by this method.



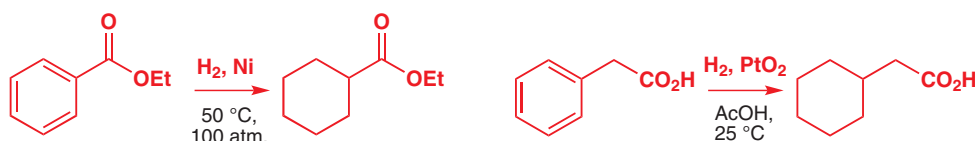
➡ See Chapter 22 for a discussion of the reactivity of α,β -unsaturated carbonyl compounds.

But what if you want instead to reduce the $\text{C}=\text{O}$ group selectively? You should immediately think of using NaBH_4 . But in Chapter 22 we pointed out that hydride reducing agents in general are not good choices for the selective reduction of the $\text{C}=\text{O}$ bond of unsaturated carbonyl compounds because they tend to add to the double bond as well, giving first the saturated carbonyl compound, which is then reduced to the alcohol. The way to get regioselective addition directly to the carbonyl group is to use NaBH_4 in the presence of a hard, Lewis-acidic metal salt, such as CeCl_3 . This combination of reagents is known as the **Luche reduction**.



How to reduce benzene rings to cyclohexanes

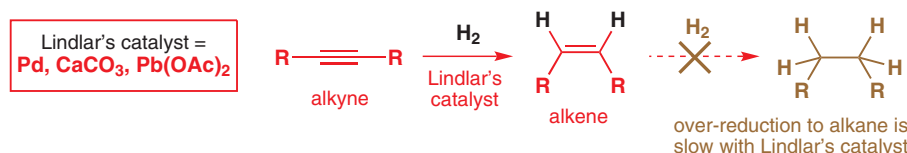
Even aromatic rings can be hydrogenated in preference to C=O groups: in these examples the ester and acid survive while phenyl is reduced to a cyclohexyl group.



The catalyst used in each reduction can be a matter of trial and error, and it is difficult to predict which metal will be most successful—generally Pt, Rh, or Ni is used for arenes.

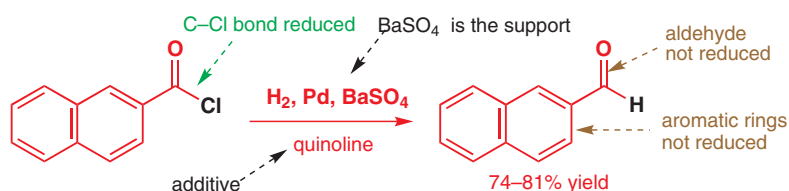
How to reduce alkynes to alkenes

A catalyst known as **Lindlar's catalyst** (which you will meet again in Chapter 27) is used to reduce alkynes to alkenes, but does not easily reduce alkenes to alkanes. This requires rather subtle chemoselectivity: alkenes are usually hydrogenated at least as fast as alkynes, so we need to be sure the reaction stops once the alkene has been formed. The Lindlar catalyst is a palladium catalyst (Pd/CaCO₃) deliberately poisoned with lead. The lead lessens the activity of the catalyst and makes further reduction of the alkene product slow: most palladium catalysts would reduce alkynes all the way to alkanes. Best selectivities are obtained if quinoline is also added to the reaction, and alkyne to alkene reductions work with Pd/BaSO₄ + quinoline too. Even so, Lindlar reactions often have to be monitored carefully to make sure that over-reduction is not taking place.



How to reduce acid chlorides to aldehydes

Catalytic hydrogenation is often chosen as a method for reduction because of its chemoselectivity for C=C over C=O groups, and an important hydrogenation involving a carbonyl compound is not actually a reduction of the C=O double bond. Hydrogenation of acyl chlorides gives aldehydes in a reaction known as the **Rosenmund reaction**—really a hydrogenolysis of a C–Cl bond.

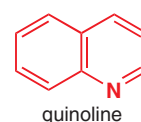


This is a good way of reducing compounds at the carboxylic acid oxidation level to aldehydes, which is why we included it in the table of carbonyl reductions on p. 534.

Hydrogenation sometimes requires high pressures of hydrogen—100 atmospheres in the reaction on the left. These reactions are carried out in a sealed apparatus known as a Parr hydrogenator, or using hydrogen generated electrolytically and pumped at high pressure within a flow system.

Raney nickel

Nickel is often used for hydrogenation in a finely divided form known as Raney nickel. Raney nickel is made from a nickel–aluminium alloy. The aluminium is dissolved away using concentrated aqueous sodium hydroxide, leaving the nickel as a fine powder. The process liberates H₂, and some of this hydrogen remains adsorbed on the nickel catalyst. This means that some hydrogenations, particularly those of C–S bonds, which you will come across later in this chapter, can be carried out just by using freshly prepared Raney nickel, with no added H₂. Raney nickel is abbreviated to RaNi—nothing to do with radium, note!

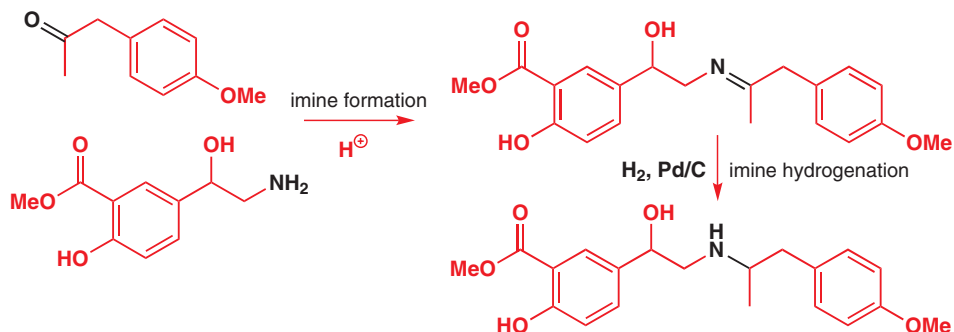


■ You will notice too that the catalyst support is different: Pd/BaSO₄ rather than Pd/C. BaSO₄ (and CaCO₃) are commonly used as supports with more easily reduced substrates because they allow the products to escape from the catalyst more rapidly and cut down over-reduction.

The quinoline is needed both to neutralize the HCl produced in the reaction and to moderate the activity of the catalyst, preventing over-reduction.

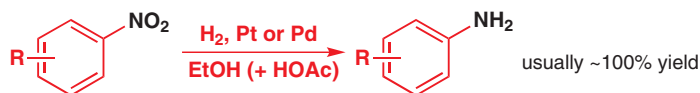
Reductive amination by catalytic hydrogenation

The unreactivity of carbonyl groups towards catalytic hydrogenation allows us to use hydrogenation in a similar way to sodium cyanoborohydride to carry out reductive aminations of amines and carbonyl compounds. For example, in the synthesis of salmefamol we presented on p. 530, one step involves the formation of the imine from an amine and a ketone in the presence of acid, hydrogen, and a palladium catalyst. The imine (in its protonated iminium form) is hydrogenated to yield an amine, while the ketone (and the aromatic systems) remains untouched.



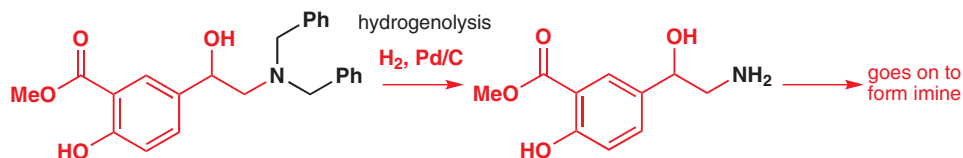
How to reduce nitro groups to amines

In Chapters 21 and 22 we saw how the sequence of nitration of aromatic rings followed by reduction was a useful route to aromatic amines. The reduction of the nitro group can be carried out by Sn/HCl but catalytic hydrogenation is much simpler. The reaction is usually done in ethanol with a Pd or Pt catalyst, and it may be necessary to add a weak acid to prevent the amine produced from poisoning the catalyst. The real gain over the Sn/HCl method is in the work-up. Instead of separating and disposing of voluminous toxic tin residues, a simple filtration to remove the catalyst, evaporation, and crystallization or distillation gives the amine.



Hydrogenolysis: breaking C–O and C–N bonds

In the reductive amination above we skirted over the fact that the starting amine in the synthesis of salmefamol (look back at p. 530) in fact carries two benzyl groups, which disappear during the hydrogenation.



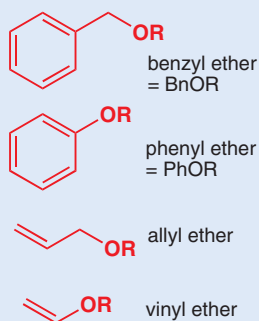
What happens to them is a **hydrogenolysis**—a reaction that is liable to occur under catalytic hydrogenation conditions whenever a heteroatom (in particular O or N) finds itself bonded to a carbon atom *adjacent* to a benzene ring, in other words with benzylic amines, alcohols or ethers.

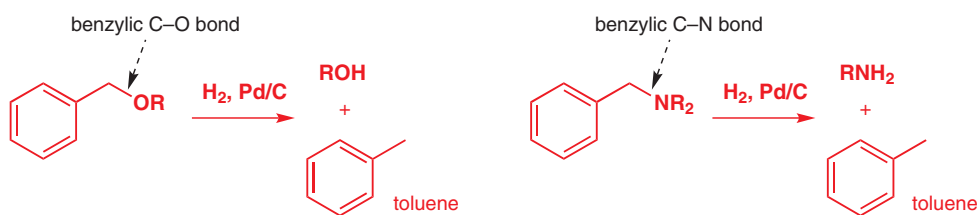
➡ You met reductive amination with sodium cyanoborohydride in Chapter 11, p. 234.

➡ For the use of nitroarenes in the synthesis of aromatic compounds, see pp. 521 and 576.

Benzyl and allyl

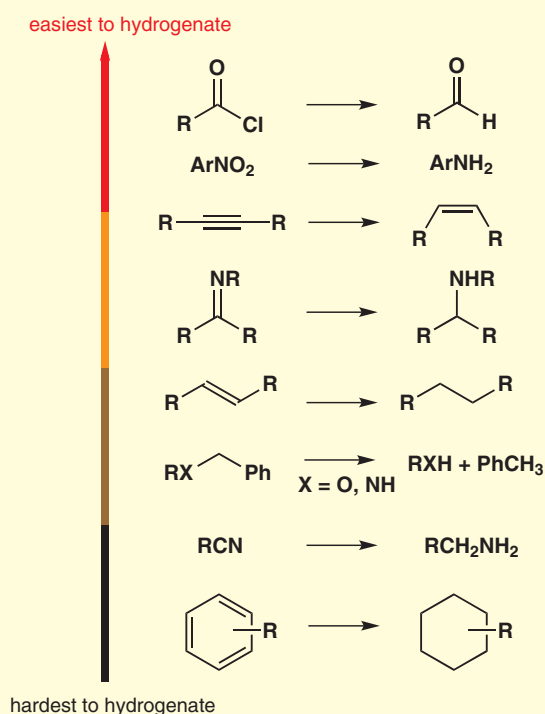
Here is a good place to remind you of the difference between a benzyl or allyl group on one hand, which are attached via an sp^3 C atom, and a phenyl or vinyl group on the other, where the bond is directly to an sp^2 C atom. You first met these groups in Chapter 2, p. 42.





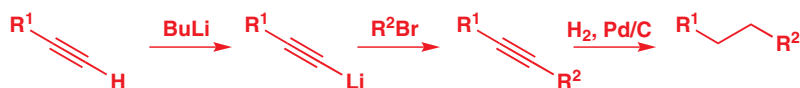
Hydrogenolysis happens under similar conditions to alkene hydrogenation, but involves breakage of a C–O or C–N σ bond rather than a C=C π bond. It is particularly important for removing benzyl protecting groups, to which we will return later in the chapter.

- We can draw up a sequence of reactivity towards hydrogenation. The precise ordering varies with the catalyst, and some catalysts are particularly selective towards certain classes of compound—for example, Pt, Rh, and Ru will selectively hydrogenate aromatic rings in the presence of benzylic C–O bonds, while with Pd catalysts the benzylic C–O bonds are reduced faster.



Getting rid of functional groups

Functional groups can be useful for putting a molecule together, but they aren't always needed in the final product. We need ways of getting rid of them. Hydrogenation of alkenes is one way that you have seen. Hydrogenation of alkynes to alkanes is very useful because we can build long chains of carbon atoms by alkylating alkynes, then hide the evidence by hydrogenation:



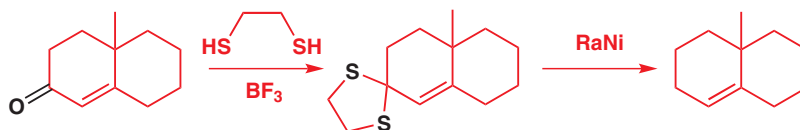
■ The hydrogenation uses Pd/C, not the Lindlar catalyst of course, because we want to go all the way to the alkane.

Alcohols can be got rid of either by elimination to alkenes and then hydrogenation or by tosylation and substitution using borohydride to provide a nucleophilic hydrogen atom.

■ Lithium triethylborohydride is used here—it is particularly good at these S_N2 substitutions—but other powerful hydride reducing agents work too.

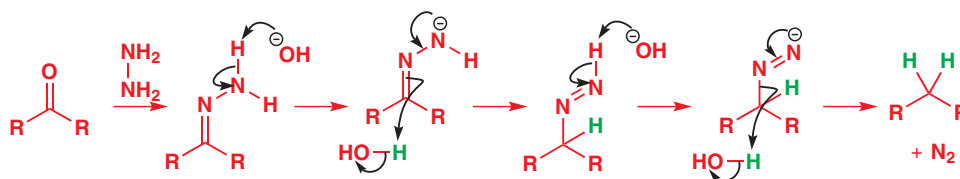


Removal of carbonyl groups is harder, although there are several possible methods. C–O bonds are strong, but C–S bonds are much weaker and are often easily reduced with Raney nickel. We can get rid of aldehyde and ketone carbonyl groups by making them into **thioacetals**, sulfur analogues of acetals, formed in a reaction analogous to acetal formation (see Chapter 11) but using a dithiol with a Lewis acid catalyst. Freshly prepared Raney nickel carries enough H_2 (p. 537) to reduce the thioacetal without added hydrogen.

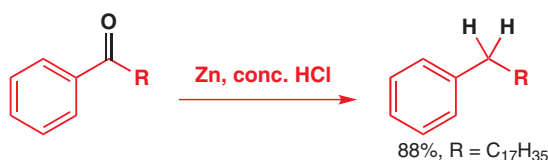


A slightly more vigorous method, known as the **Wolff–Kishner reduction**, is driven by the elimination of nitrogen gas from a hydrazone. Hot concentrated sodium hydroxide solution deprotonates the hydrazone, which can then lose nitrogen to form an alkyl anion, which is immediately protonated by water.

Interactive mechanism for Wolff–Kishner reduction



The third method is the simplest to do, but has the most complicated mechanism. The **Clemmensen reduction** is also rather violent, and really reasonable only for compounds with just the one functional group. It uses zinc metal dissolving in concentrated hydrochloric acid. As the metal dissolves, it gives up two electrons—in the absence of something else to do, these electrons would reduce the H^+ in the acid to H_2 , and give ZnCl_2 and H_2 . But in the presence of a carbonyl compound, the electrons go to reduce the C=O bond.



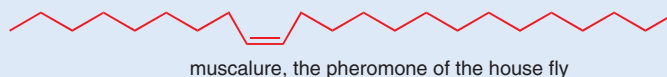
■ You saw this reaction in Chapter 21 as a useful way of turning acylated arenes of the type that are easy to make by Friedel–Crafts acylation into alkylated arenes of the type that are hard to make by Friedel–Crafts alkylation.

The mechanism has a good deal in common with a whole class of reductions, of which the Clemmensen is a member, known as **dissolving metal reductions**. We shall now look at these as our third (after metal hydrides and catalytic hydrogenation) important class of reducing agents.

Two synthetic routes to muscalure—the house fly pheromone

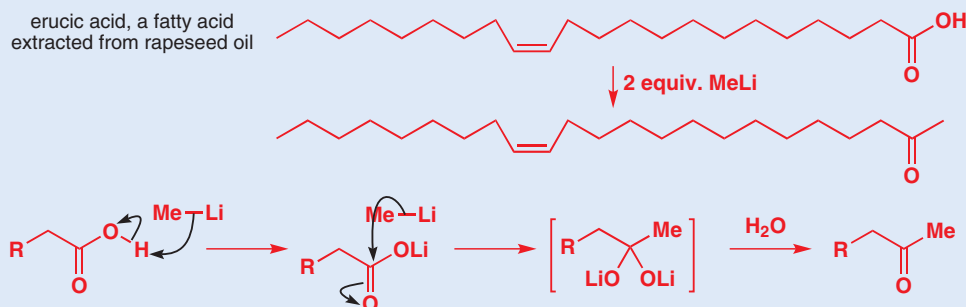
Many insects attract a mate by releasing a volatile organic compound known as a pheromone. Pheromones are highly specific to species and provide a cunning means of controlling pests: place a pad of cotton wool soaked in male pheromone inside a trap and in drop all the female pests—no next generation. If insect control is to rely on a supply of the pheromone, that supply has to be synthetic—it takes enormous numbers of squashed insects to provide even a few milligrams of most pheromones.

Two syntheses of the very simple pheromone of a very common insect—the house fly—provide an illustration of how to use two of the reduction methods we have just described. The pheromone, known as muscalure, is a Z-alkene.



One approach, used by some American chemists in the early 1970s, was very simple. These chemists noted the similarity between the structures of muscalure and the fatty acid known as erucic acid, which is abundant in rapeseed oil, and decided to make muscalure from erucic acid. They first reacted the acid with

two equivalents of methyllithium—the first equivalent deprotonates the acid to make a lithium carboxylate salt, while the second reacts with the lithium carboxylate to make a ketone (see p. 219).



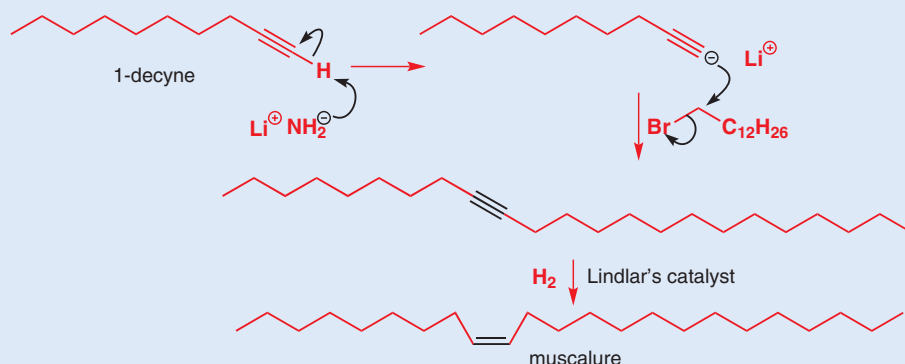
The next step is to remove the ketone functional group. The method chosen was the Wolff–Kishner reaction described on p. 540: make a hydrazone and

heat in the presence of base. Muscalure is the product.



Later some Russian chemists made the same compound by a different route. They chose to introduce the *Z* double bond by hydrogenation of an alkyne over Lindlar's catalyst (p. 537). To make the alkyne they needed, they took 1-decyne,

treated it with LiNH_2 to remove the acidic terminal proton, and reacted the anion with an *n*-alkyl bromide. By stirring the alkyne with Lindlar's catalyst under an atmosphere of hydrogen they were able to make muscalure.

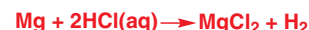


Dissolving metal reductions

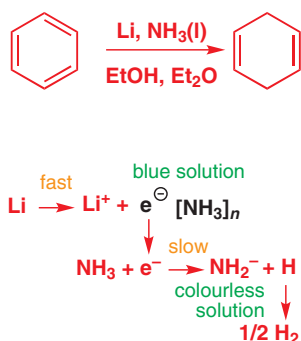
You will be familiar with the idea that many metals react with acid to liberate hydrogen, forming a salt at the same time. There is an example in the margin. The metal cation (Mg^{2+} in this example) results from the loss of electrons, and these electrons reduce $2 \times \text{H}^+$ to give H_2 .

The same thing happens even in very weak acids (water, alcohols...even liquid ammonia) if the metal is very reactive (sodium or potassium, say). You can think of the process here in two steps: first sodium releases an electron, then the electron is captured by a proton from NH_3 to give H , which forms H_2 . Sodium ethoxide (NaOEt) and sodium amide (NaNH_2 , p. 171) are made by dissolving sodium in ethanol or liquid ammonia, respectively.

But what if, instead of just reducing the solvent to liberate hydrogen, we harness the electrons by giving them a more easily reduced substrate instead? A dissolving metal reduction results: note *dissolving*. The electrons have to be captured as the metal releases them, otherwise they will just reduce the solvent to give H_2 .



■ You have in fact already met several dissolving metal reductions, for example Sn, HCl for reducing nitro groups (p. 495) and the Clemmensen reduction mentioned above.



■ Sodium amide, NaNH₂, the base you met early in this book, is made by dissolving Na in liquid NH₃ and then waiting till the solution is no longer blue.

The regiochemistry of the final product is determined at the last protonation step—the anion itself is of course delocalized and could react at either end to give a conjugated diene, which would be more stable. Why then does it choose to pick up a proton in the middle and give a less stable isomer? Kinetically controlled reactions of pentadienyl anions with electrophiles typically take place at this central carbon as a result of orbital interactions. For more information see the further reading section at the end of the chapter.

➡ The terms *ipso*, *ortho*, *meta*, and *para* were defined in Chapter 18, p. 416.

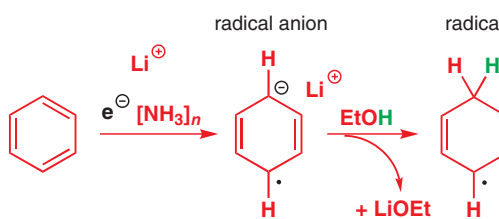
Dissolving metal reductions work because the electrons released as reactive metals form soluble cations that can be harnessed to do other, more useful, reductions. Electrons are the simplest possible reducing agents, and they will reduce carbonyl compounds, alkynes, or aromatic rings—in fact any functional group with a low-energy π orbital into which the electron can go.

Birch reduction of arenes

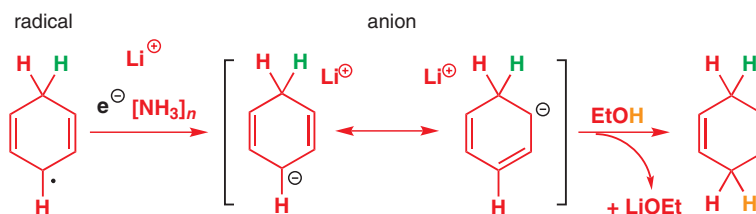
We shall start by looking at the dissolving metal reduction of aromatic rings, known as the **Birch reduction**. The margin shows the reaction of benzene with lithium in liquid ammonia. At first sight this reaction looks improbable, with an aromatic ring ending up as an unconjugated diene. The mechanism will explain why we get this regiochemistry, and also why the reaction stops there—in other words, why the dissolving lithium reduces an aromatic ring more readily than an alkene.

The first thing to note is that when lithium or sodium dissolve in ammonia they give an intense blue solution. Blue is the colour of solvated electrons: these group 1 metals ionize to give Li⁺ or Na⁺ and e⁻(NH₃)_n. With time, the blue colour fades, as the electrons reduce the ammonia to NH₂⁻ and H₂.

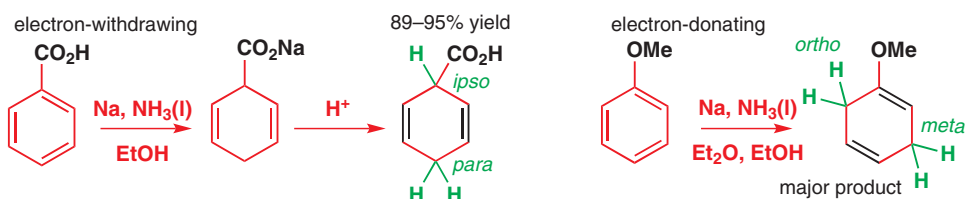
Birch reductions use those blue solutions, with their solvated electrons, as reducing agents. The reduction of NH₃ to NH₂⁻ and H₂ is quite slow, and a better electron acceptor will get reduced in preference. With benzene, the electrons go into the lowest lying antibonding orbital (benzene's LUMO). The species we get can be represented in several ways, all of them radical anions (molecules with one excess unpaired electron). The radical anion is very basic, and it picks up a proton from the ethanol that is in the reaction mixture.



The molecule is now no longer anionic, but it is still a radical. It can pick up another electron, which pairs with the radical to give an anion, which is quenched again by the proton source (ethanol). Overall we have arrived at two additional H atoms by sequentially adding two electrons plus two protons.



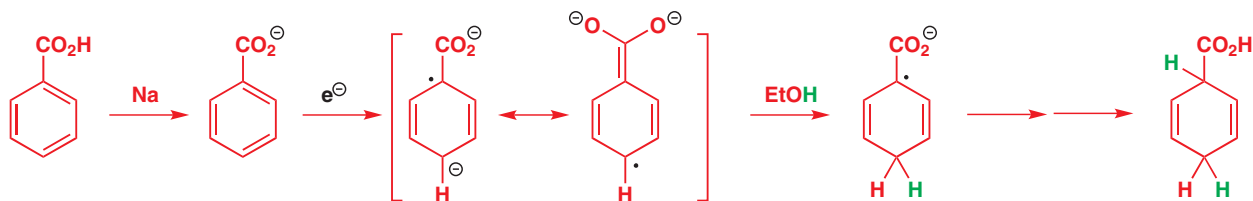
More questions of regioselectivity arise when there are substituents around the aromatic ring. Here are two examples.



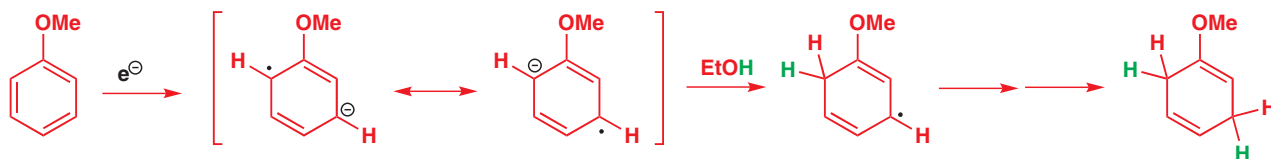
These examples serve to illustrate a general principle:

- Electron-withdrawing groups promote *ipso*, *para* Birch reduction.
- Electron-donating groups promote *ortho*, *meta* Birch reduction.

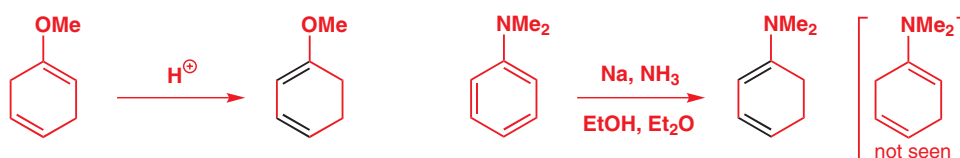
The explanation must lie in the distribution of electron density in the intermediate radical anions. Electron-withdrawing groups stabilize electron density at the *ipso* and *para* positions, and protonation occurs *para*,



while electron-donating groups stabilize *ortho* and *meta* electron density:



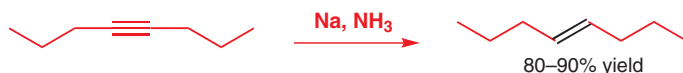
If you want the conjugated dienes as products, it is quite a simple matter to isomerize them using an acid catalyst. In fact, a small amount (about 20%) of the conjugated product is produced anyway in the reaction of anisole above. With anilines, it is impossible to stop the isomerization taking place during the reaction, and Birch reduction always gives conjugated dienamines.



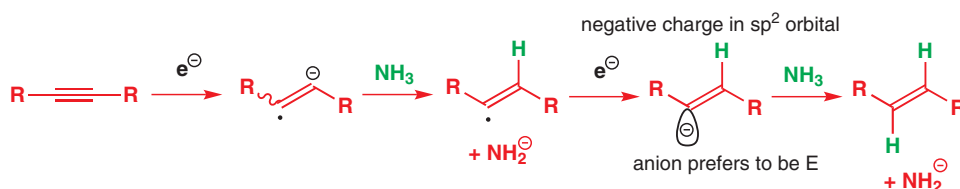
■ Make sure you can write a mechanism for the isomerization into conjugation. *Hint.* Start as though you were protonating an enol ether on carbon. You saw this sort of thing in Chapter 20.

Birch reduction of alkynes

Birch reduction works for alkynes too and reduces them to *trans* alkenes.



The mechanism follows the same course as the reduction of aromatic rings, but the vinyl anion is basic enough to deprotonate ammonia, so no added proton source is required. Vinyl anions are geometrically unstable, and choose to be *E*. Again, the two green H atoms come from two electrons and two protons.

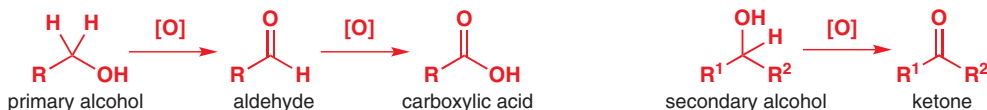


■ Note that the stereoselectivity is complementary to that of hydrogenation over the Lindlar catalyst, p. 537.

Selectivity in oxidation reactions

➡ Turn back to p. 194 (at the end of Chapter 9) if you need reminding of this.

■ [O] means an unspecified oxidizing agent.



Since then you have met some other oxidizing agents too, particularly in Chapter 19:

- On p. 429 you saw peracids as oxidizing agents for C=C double bonds—they give epoxides.
- On p. 442 you saw osmium tetroxide (OsO_4) giving diols from alkenes.
- On p. 443 we introduced ozone (O_3) as a means of cleaving alkenes to carbonyl compounds by ozonolysis.

Unlike Cr(VI), none of these reagents will oxidize a hydroxyl group: they are *chemoselective* for C=C double bonds, but do not react with hydroxyl groups. By contrast, Cr(VI) oxidizes alcohols but not alkenes.

Oxidizing agents

Chemoselective for C=C double bonds

peracids, RCO_3H (Chapter 19)
osmium tetroxide, OsO_4 (Chapters 19 and 34)
ozone, O_3 (Chapters 19 and 34)

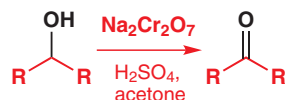
Chemoselective for alcohols or carbonyl compounds

Cr(VI) compounds
Mn(VII) compounds
some high oxidation state Hal, N, or S compounds

In this section we will be concerned only with oxidizing agents that oxidize alcohols and carbonyl compounds, and in particular we shall be concerned with ways of choosing whether to arrest the oxidation of primary alcohols at the aldehyde stage or let it continue to the carboxylic acid.

The most commonly used methods for oxidizing alcohols are based around metals in high oxidation states, often chromium(VI) (which you met in Chapter 9) or manganese(VII), and you will see that mechanistically they are quite similar—they both rely on the formation of a bond between the hydroxyl group and the metal. Another class of oxidations, those that use halogens, sulfur, or nitrogen in high oxidation states, we will deal with relatively briefly.

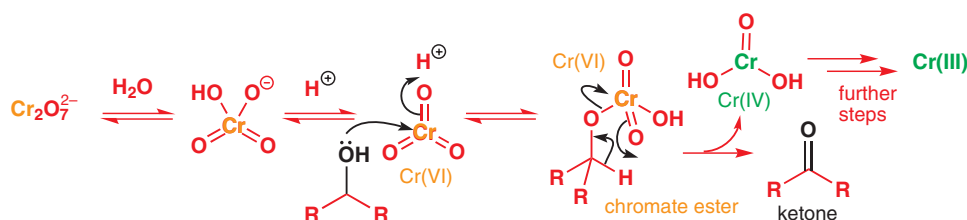
How to oxidize secondary alcohols to ketones



🖱 Interactive mechanism for chromium(VI) oxidation of alcohols

➡ We discussed this mechanism on p. 195.

You met methods for this reaction in Chapter 9, where you met the use of Cr(VI) in the form of CrO_3 . One common version of this reaction is the *Jones oxidation*, shown in the margin. The mechanism starts with the formation of HCrO_4^- ions, that is, Cr(VI), from dichromate ion in solution. In acid, these Cr(VI) species form chromate esters with alcohols. Chromate esters decompose by elimination of a Cr(IV) species, which subsequently reacts with Cr(VI) to yield $2 \times \text{Cr(V)}$. These Cr(V) species can oxidize alcohols in the same way and are thereby reduced to Cr(III) (the final metal-containing by-product). Cr(VI) is orange and Cr(III) is green, so the progress of the reaction is easy to follow by colour change.

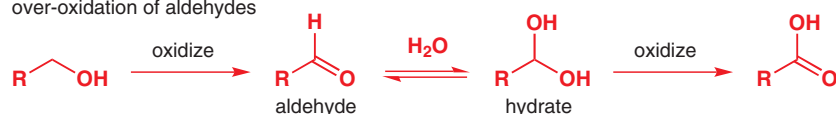


Chromic acid is best avoided if acid-sensitive alcohols are to be oxidized, and an alternative reagent for these is PCC (pyridinium chlorochromate), which can be used in dichloromethane.

How to oxidize primary alcohols to aldehydes

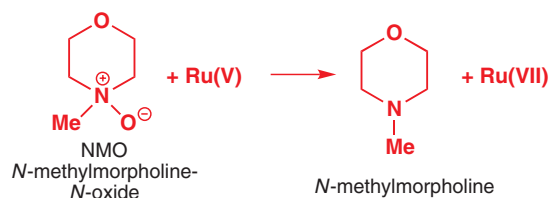
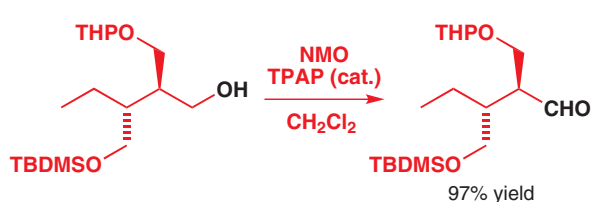
Aqueous methods like the Jones oxidation are no good for this, since the aldehyde that forms is further oxidized to acid via its hydrate. The oxidizing agent treats the hydrate as an alcohol, and oxidizes it to the acid.

over-oxidation of aldehydes



The key thing is to avoid water, so PCC in dichloromethane works quite well. The related reagent PDC (pyridinium dichromate) is particularly suitable for oxidation to aldehydes.

Some very mild oxidizing agents are being more and more widely used for the synthesis of very sensitive aldehydes. One of these is known as TPAP (tetra-*n*-propylammonium perruthenate, pronounced 'tee-pap'). TPAP can be used catalytically, avoiding the large amounts of toxic heavy metal by-products generated by most chromium oxidations. The stoichiometric oxidant in this reaction is NMO (*N*-methylmorpholine-*N*-oxide), which is reduced to the amine, reoxidizing the ruthenium to Ru(VII).

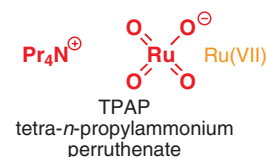
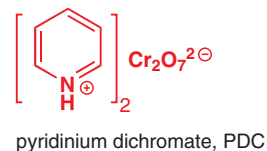
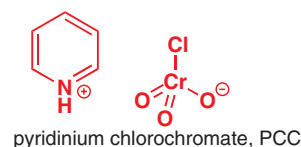
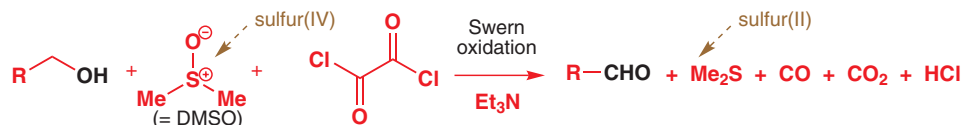


Another important mild oxidant is a high-valent iodine compound known as the **Dess–Martin periodinane**. It can be made from 2-iodobenzoic acid.



It will oxidize even very sensitive alcohols to carbonyl compounds—few others, for example, would give the *cis*- α,β -unsaturated aldehyde in the margin from a *cis*-allylic alcohol without isomerizing it to *trans*, or producing other by-products.

We shall leave detailed discussion of one more method till much later, in Chapter 27, since the mechanism involves some sulfur chemistry you will meet there, but we introduce it here because of its synthetic importance. Known as the **Swern oxidation**, it uses a sulfoxide [sulfur(IV)] as the oxidizing agent. The sulfoxide is reduced to a sulfide, while the alcohol is oxidized to an aldehyde.



➡ The abbreviations THP and TBDMS will be explained later in this chapter (p. 550).

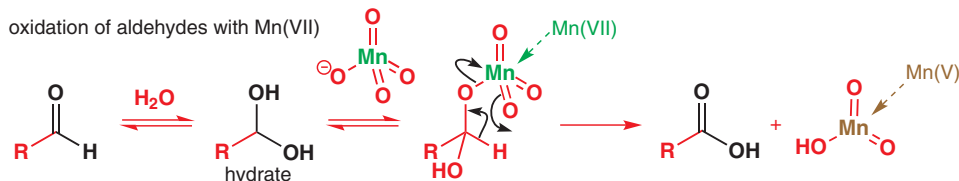
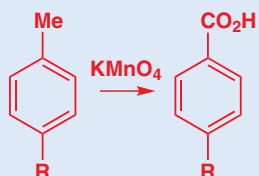


➡ For more details, and the mechanism of the Swern oxidation, see p. 667.

How to oxidize primary alcohols or aldehydes to carboxylic acids

Sometimes the ‘over-oxidation’ we were trying to avoid in oxidizing alcohols to aldehydes is actually the reaction you want. It’s best done with an aqueous solution of Cr(VI) or Mn(VII). Acidic or basic aqueous potassium permanganate is often a good choice. From alcohols in acidic solution the mechanism follows very much the lines of the chromic acid mechanism; from aldehydes, the mechanism is very similar.

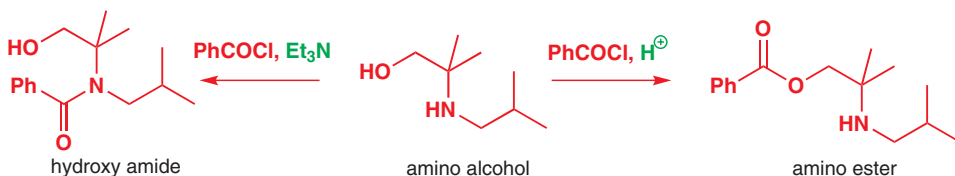
Potassium permanganate is a very powerful oxidant and will also oxidize a benzylic methyl group (i.e. a toluene derivative) to a carboxylic acid. You saw this in one of the steps in the synthesis of saccharin on p. 486.



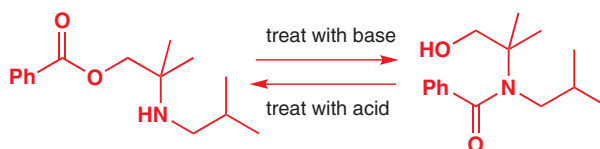
Competing reactivity: choosing which group reacts

We hope that our survey of the important methods for reduction and oxidation has shown you that, by choosing the right reagent, you can often get reaction only at the functional group you want. The chemoselectivity you obtain is *kinetic chemoselectivity*—reaction at one functional group is simply faster than at another.

Now look at the acylation of an amino alcohol (which is, in fact, a synthesis of the painkiller isobucaine) using benzoyl chloride under *acid* conditions. The hydroxyl group is acylated to form an ester. Yet under *basic* conditions, the selectivity is quite different, and an amide is formed.

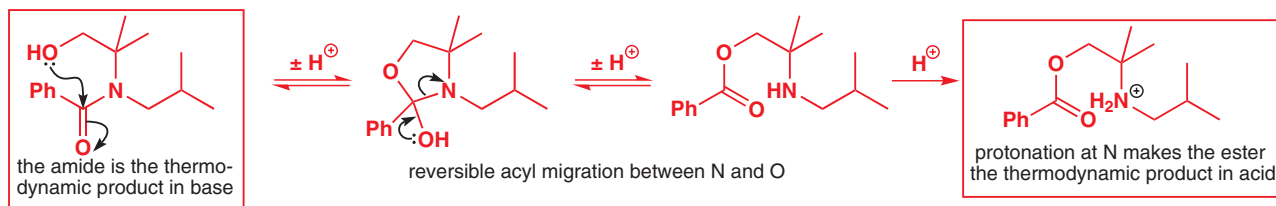


A clue to why the selectivity reverses is shown below—it is, in fact, possible to interconvert the ester and the amide simply by treating either with acid or with base.



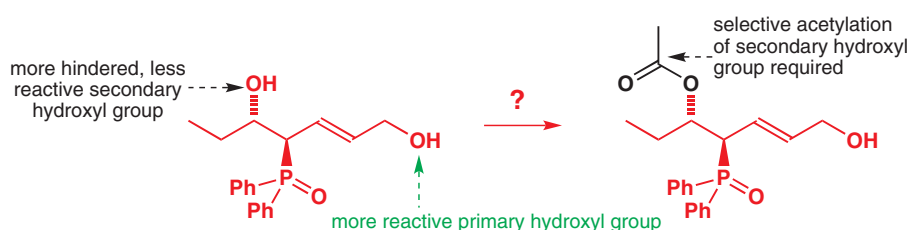
We discussed kinetic and thermodynamic control in Chapter 12 (p. 264) and again on pp. 435 and 505.

The selectivity in these reactions is *thermodynamic* chemoselectivity. Under conditions in which the ester and amide can equilibrate, the product obtained is the more stable of the two, not necessarily the one that is formed faster. In base the more stable amide predominates, while in acid the amine is protonated, which prevents it from acting as a nucleophile and removes it from the equilibrium, giving the ester.

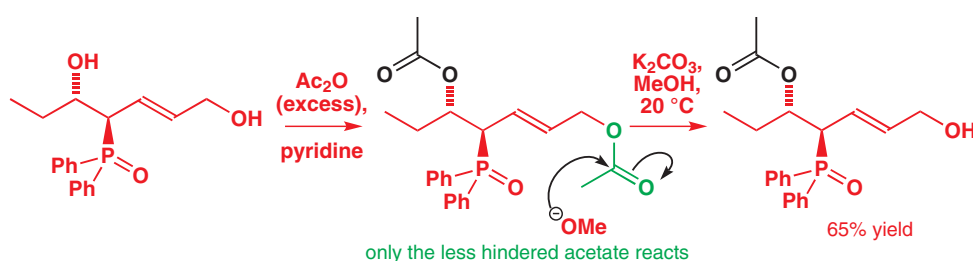


How to react the less reactive group (I): react both then 'unreact' one

The relative reactivity of the alcohol and amine in the example just given could be overturned by conducting a reaction under thermodynamic control. In kinetically controlled reactions, the idea that you can conduct chemoselective reactions on the more reactive of a pair of functional groups—carbonyl-based ones, for example—is straightforward. But what if you want to react the less reactive of the pair? There are two commonly used solutions. The first is illustrated by a compound needed by chemists in Cambridge to study an epoxidation reaction. They were able to make the following diol, but wanted to acetylate only the more hindered secondary hydroxyl group.



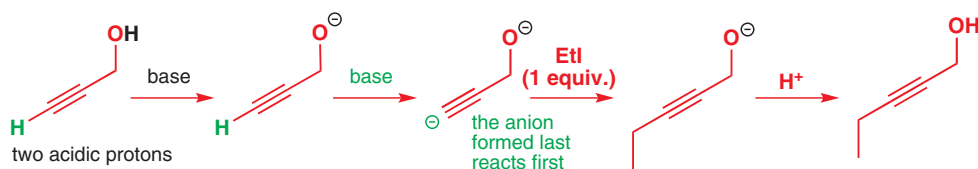
Treatment with one equivalent of an acetylating agent is no good because the primary hydroxyl group is more reactive; instead, the chemists acetylated both hydroxyl groups, and then treated the bis-acetate with mildly basic methanol (K_2CO_3 , MeOH, 20°C), which reacted only at the less hindered acetoxy group and gave the desired compound in 65% yield.



In other words, start by letting both groups react and then go backwards so the reaction is reversed, but at only one of the two groups. Steric hindrance means that the less favourable reaction (in other words, reaction at the less reactive group) was also less readily reversed.

Chemoselectivity in the reactions of dianions

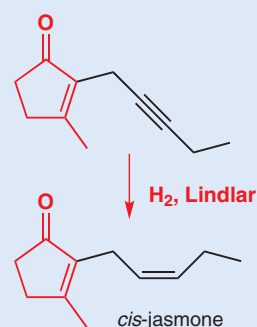
A similar idea is central to a useful bit of chemoselectivity that can be obtained in the reactions of dianions. 1-Propynol can be deprotonated twice by strong bases—first, at the hydroxyl group to make an alkoxide anion (the pK_a of the OH group is about 16) and, secondly, at the alkyne (pK_a of the order of 25) to make a dianion. When this dianion reacts with electrophiles it always reacts at the alkynyl anion and not at the alkoxide.



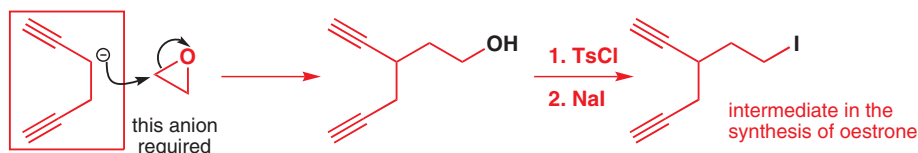
• Reactivity of dianions

The anion that is formed *last* reacts *first*.

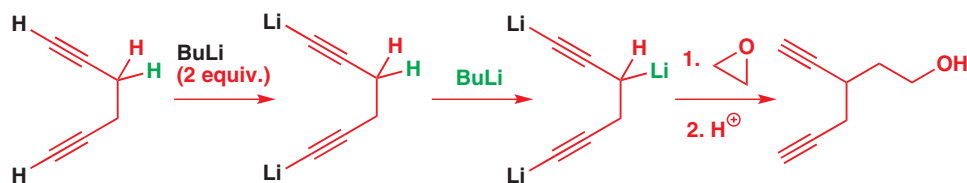
This reaction is important in a synthesis of the perfumery compound *cis*-jasmone. The alkyne is the precursor to *cis*-jasmone's alkene side chain.



Vollhardt used this sort of chemoselectivity in his 1977 synthesis of the female sex hormone oestrone. He needed an alkyl iodide, which could be made by reacting an anion of a bis-alkyne with ethylene oxide.

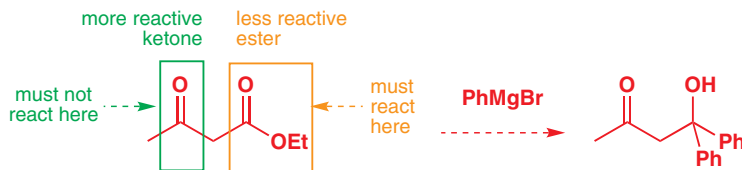


Although anions can often be formed straightforwardly next to alkynes, there are two other more acidic protons (black) in the molecule that would be removed by base before the green proton. However, treatment with *three* equivalents of butyl lithium removes all three, and the trianion reacts with ethylene oxide at the last-formed anionic centre to give the required compound.

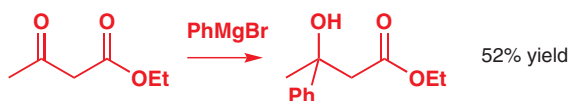


How to react the less reactive group (II): protecting groups

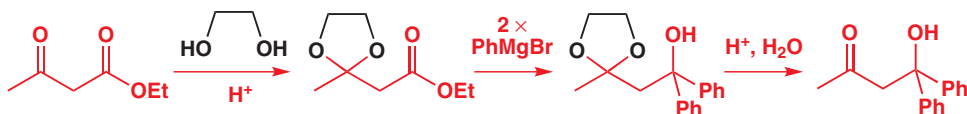
The more usual way of reacting a less reactive group in the presence of a more reactive one is to use a **protecting group**. This tertiary alcohol, for example, could be made from a keto-ester if we could get phenylmagnesium bromide to react with the ester rather than with the ketone.



As you would expect, simply adding phenylmagnesium bromide to ethyl acetoacetate leads mainly to addition to the more electrophilic ketone.



One way of making the alcohol we want is to *protect* the ketone from attack by disguising it from the nucleophile. An acetal protecting group (shown in black) is used.



■ Five-membered cyclic acetals like these are known as dioxolanes. You met them first in Chapter 11 when we were discussing acetal formation and hydrolysis.

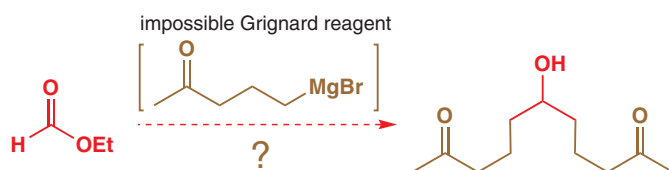
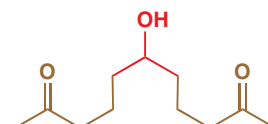
The first step puts the protecting group on to the (more electrophilic) ketone carbonyl, making it no longer reactive towards nucleophilic addition. The Grignard then adds to the ester, and finally a 'deprotection' step, acid-catalysed hydrolysis of the acetal, gives us back the ketone. An acetal is an ideal choice here—acetals are stable to base (the conditions of the reaction we want to do), but are readily cleaved in acid.

A survey of protecting groups

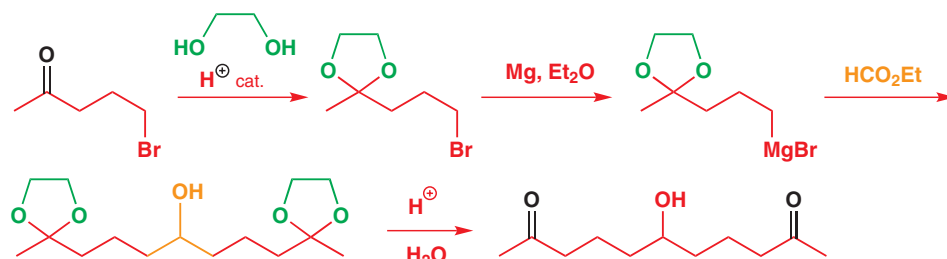
A dioxolane can be used in this way to protect aldehydes and ketones from powerful, basic nucleophiles, and makes the first entry in the tour of important protecting groups we shall conduct you through in the next few pages.

Protecting group	Structure	Protects	From	To protect	To deprotect
acetal (dioxolane)		ketones, aldehydes	nucleophiles, bases		H^+ , H_2O

By protecting sensitive functional groups like ketones it becomes possible to make reagents that would otherwise be unstable. In a synthesis of the natural product porantherine, a compound based on the structure in the margin was needed. As it's a symmetrical secondary alcohol (see p. 216), a good way to make it is to add a Grignard reagent twice to ethyl formate.

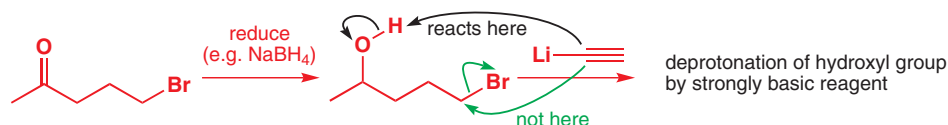


But, of course, a ketone-containing Grignard is an impossibility as it would self-destruct, so an acetal-protected compound was used. Acid-catalysed hydrolysis of the two dioxolanes, coloured green, reveals the diketone.



Strongly nucleophilic reagents like Grignard reagents and organolithiums are also strong bases and may need protecting from acidic protons as well as from electrophilic carbonyl groups. Among the most troublesome are the protons of hydroxyl groups. When some American chemists wanted to make the antiviral agent Brefeldin A, they needed the simple alkynol in the margin.

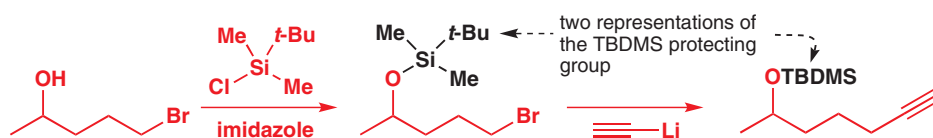
A synthesis could start with the same bromoketone as the one above: reduction gives an alcohol, but alkylation of an alkynyl anion with this compound is not possible because the anion will just deprotonate the hydroxyl group.



imidazole = (a weak base)

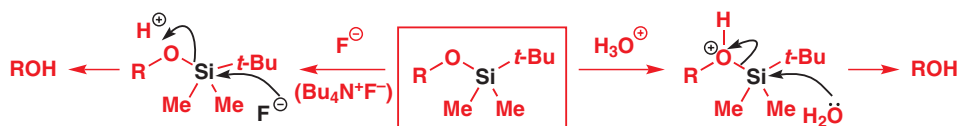
The answer is to protect the hydroxyl group with a group resistant to base, and the group chosen here was a **silyl ether**. Such ethers are made by reacting the alcohol with a trialkylsilyl chloride (here *tert*-butyldimethylsilyl chloride, or TBDMSCl) in the presence of a weak base, usually imidazole, which also acts as a nucleophilic catalyst (Chapter 12).

➔ You met imidazole on p. 178.



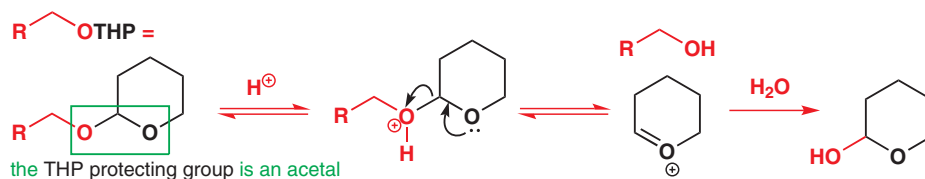
Silicon has a strong affinity for electronegative elements, particularly O, F, and Cl, so trialkylsilyl ethers are attacked by hydroxide ion, water, or fluoride ion but are more stable to carbon or nitrogen bases or nucleophiles. They are usually removed with aqueous acid or fluoride salts, particularly $\text{Bu}_4\text{N}^+\text{F}^-$ (tetra-*n*-butylammonium fluoride, known as TBAF and pronounced ‘tea-baff’), which is soluble in organic solvents. In fact, TBDMS is one member of a whole family of trialkylsilyl protecting groups and their relative stability to nucleophiles of various kinds is determined by the three alkyl groups carried by silicon. The most labile, trimethylsilyl (TMS), is removed simply on treatment with methanol, while the most stable require hydrofluoric acid.

■ Although not important to our discussion here, these substitution reactions are not the simple $\text{S}_{\text{N}}2$ reactions (Chapter 15) they might appear to be. The nucleophile adds to silicon first to form a five-valent anion, which decomposes with the loss of the alcohol.



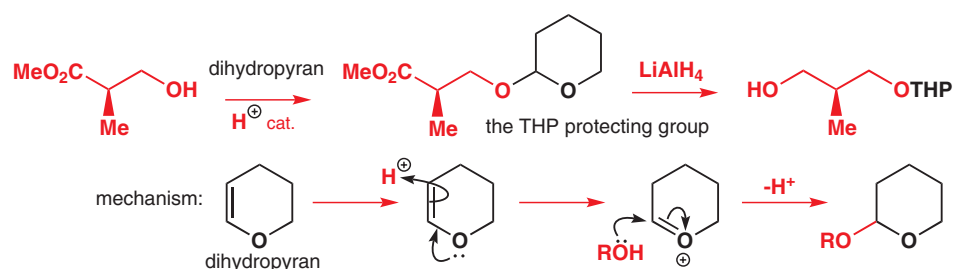
Protecting group	Structure	Protects	From	Protection	Deprotection
trialkylsilyl $\text{R}_3\text{Si}-$, e.g. TBDMS	$\text{RO}-\text{SiMe}_3$ $\text{RO}-\text{SiMe}_2\text{-}t\text{-Bu}$	alcohols (OH in general)	nucleophiles, C or N bases	R_3SiCl , base	H^+ , H_2O , or F^-

Why can't we just use a simple alkyl ether (methyl, say) to protect a hydroxyl group? There is no problem making the ether, and it will survive most reactions—but there is a problem getting an ether off again. This is always a consideration in protecting group chemistry—you want a group that is stable to the conditions of whatever reaction you are going to do (in these examples, strong bases and nucleophiles), but can then be removed under mild conditions that do not result in decomposition of a sensitive molecule. What we need then is an ether that has an Achilles' heel—a feature that makes it susceptible to attack by some specific reagent or under specific conditions. One such group is the tetrahydropyranyl (THP) group. Although it is stable under basic conditions, as an ether would be, it is an acetal—the presence of the second oxygen atom is its Achilles' heel and makes the THP protecting group susceptible to hydrolysis under acidic conditions. You could see the lone pair on the second oxygen atom as a safety catch that is released only in the presence of acid.



➡ There is more chemistry of enol ethers in Chapter 20.

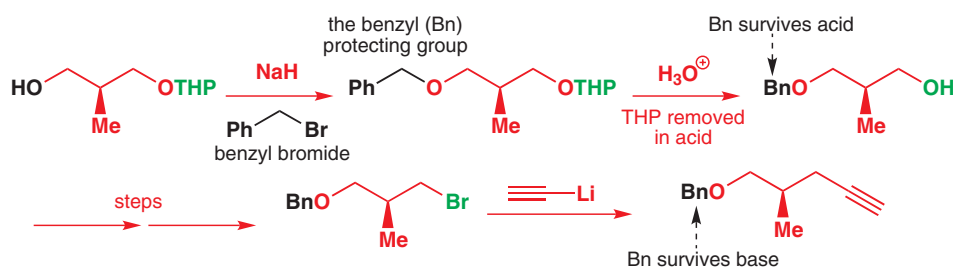
Making the THP acetal has to be done in an unfamiliar way because the usual ‘carbonyl plus two alcohols’ method is inappropriate (work out why!). Alcohols are protected by treating them with an enol ether, dihydropyran, under acid catalysis. Notice the oxonium intermediate (formed by a familiar mechanism from Chapter 12)—just as in a normal acetal-forming reaction. In this example the THP group is at work preventing a hydroxyl group from interfering in the reduction of an ester.



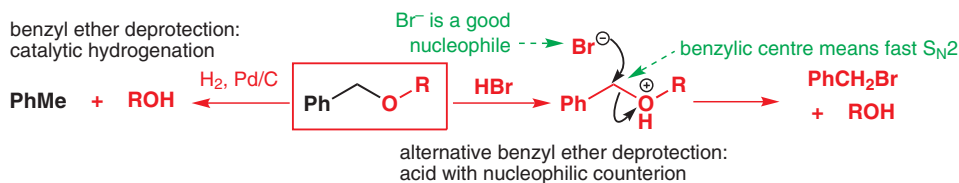
Protecting group	Structure	Protects	From	To protect	To deprotect
tetrahydropyranyl (THP)		alcohols (OH in general)	strong bases	 dihydro-pyran and acid	H^+ , H_2O

The THP-protected compound above was used as an intermediate in a synthesis of the insecticide milbemycin. It needed to be converted to the alkyne in the margin—to do this the *other* hydroxyl group also needed protecting.

This time, however, TBDMS will not do because the protecting group needs to withstand the acidic conditions needed to remove the THP protecting group! What is more, the protecting group needs to be able to survive acid conditions in later steps of the synthesis of the insecticide. The answer is to use a third type of hydroxyl-protecting group, a benzyl ether. Benzyl (Bn) protecting groups are put on using strong base (usually sodium hydride) plus benzyl bromide, and are stable to both acid and base.

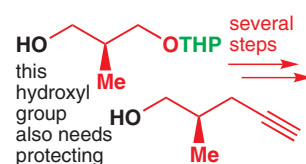


The benzyl ether's Achilles' heel is the aromatic ring and, after reading the first half of this chapter, you should be able to suggest conditions that will take it off again: hydrogenation (hydrogenolysis) over a palladium catalyst, which cleaves benzylic C–O bonds.



Benzyl ethers can alternatively sometimes be removed by acid, if the acid has a *nucleophilic* conjugate base. HBr, for example, will remove a benzyl ether because Br[−] is a good enough nucleophile to displace ROH, although only at the reactive, benzylic centre.

A little further inspection will show you that the THP group here is not just stopping the OH interfering with the LiAlH₄ reduction, but is also crucial to the preservation of the chirality of this compound. The wedged bond shows you that the starting material is a single enantiomer: without a protecting group on one of the hydroxyls, they would be identical and the compound would no longer be chiral. The THP group also complicates the situation by introducing an extra chiral centre, and hence the potential for two diastereoisomers, which we will ignore.



➔ You saw this method for making ethers by an S_N2 reaction back in Chapter 15.

Benzyl and benzoyl

Note the abbreviation for a benzyl ether, ROCH₂Ph, is ROBn. Contrast this with benzoyl esters, ROCOPh, which may be abbreviated ROBz.

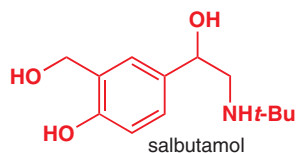
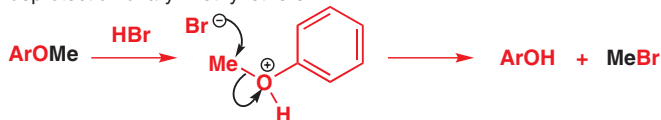
■ It must be a *palladium* catalyst—platinum would catalyse hydrogenation of the aromatic ring.

Protecting group	Structure	Protects	From	To protect	To deprotect
benzyl ether (OBn)		alcohols (OH in general)	almost everything	NaH, BnBr	H ₂ , Pd/C, or HBr
methyl ether (ArOMe)		phenols (ArOH)	bases	NaH, MeI, or (MeO) ₂ SO ₂	BBr ₃ , HBr, HI, Me ₃ SiI

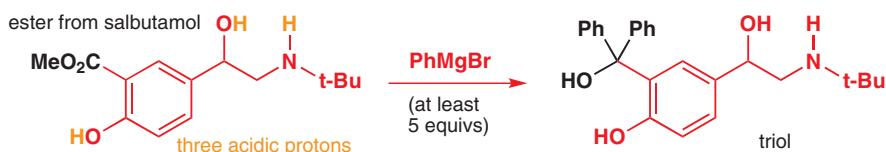
We said earlier that simple methyl ethers are inappropriate as protecting groups for OH because they are too hard to take off again. That is usually true, but not if the OH is phenolic—ArOH is a better leaving group than ROH, so HBr will take off methyl groups from aryl methyl ethers too.

■ Alternatives to HBr include BBr₃, usually the favoured reagent, HI, and Me₃SiI.

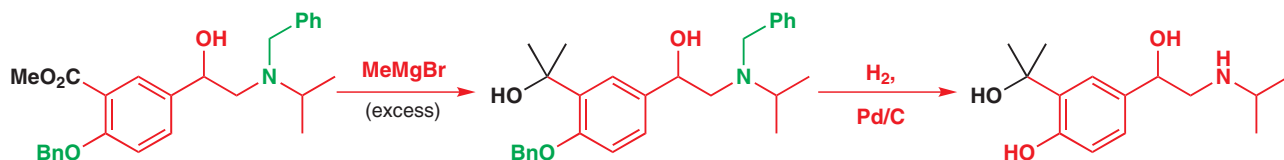
deprotection of aryl methyl ethers



Protecting groups may be useful, but they are also wasteful—both of time, because there are two extra steps to do (putting the group on and taking it off), and of material, because these steps may not go in 100% yield. Here's one way to avoid using them. During the development of the anti-asthma drug salbutamol, the triol below was needed. With large quantities of salbutamol already available, it seemed most straightforward to make the triol by adding phenylmagnesium bromide to an ester available from salbutamol. Unfortunately, the ester also contains three acidic protons, making it look as though the hydroxyl and amine groups all need protecting. But, in fact, it was possible to do the reaction just by adding a large excess of Grignard reagent: enough to remove the acidic protons *and* to add to the ester.



This strategy is easy to try, and, providing the Grignard reagent isn't valuable (you can buy PhMgBr in bottles), is much more economical than putting on protecting groups and taking them off again. But it doesn't always work—there is no way of telling whether it will until you try the reaction in the laboratory. In this closely related reaction, for example, the same chemists found that they needed to protect both the phenolic hydroxyl group (but not the other alcohol OH) as a benzyl ether and the amine NH as a benzyl amine. Both protecting groups come off in one hydrogenation step.



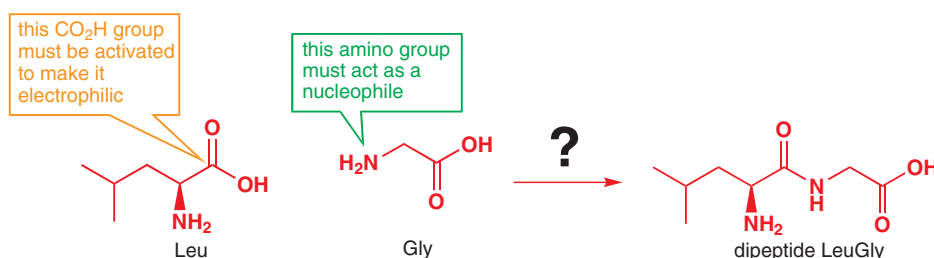
Benzyl groups are one way of protecting secondary amines against strong bases that might deprotonate them. But it is the nucleophilicity of amines that usually poses problems of chemoselectivity, rather than the acidity of their NH groups. The potential for pitfalls is

nowhere more acute than in the synthesis of one of the most important classes of biological molecules: peptides.

Peptide synthesis

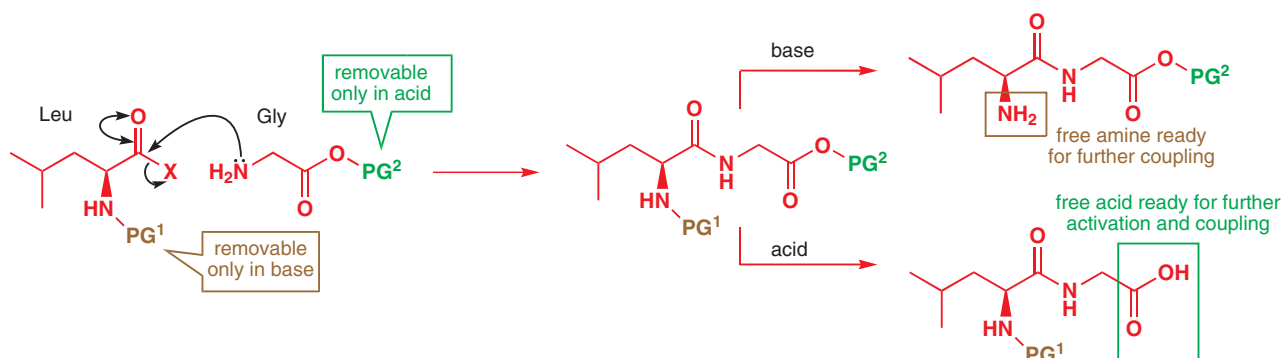
Peptide synthesis has become one of the most reliable and predictable fields of practical organic chemistry, principally because of the effectiveness of the protecting groups it employs. For this reason, peptides are one of the few classes of complex organic molecules that can routinely be made by machines, such as the one on the right, in which much of the chemistry we are about to talk about takes place without any human intervention.

Biology makes peptides and proteins by selectively coupling together members of a pool of 20 or so amino acids. To do the same in the laboratory, we need to overcome a number of challenges. For example, we'll start by thinking about how to react two amino acids together, to make a dipeptide—leucine and glycine, for example. If we want the NH_2 group of glycine to react with the CO_2H group of leucine we will first have to activate the carboxylic acid towards nucleophilic substitution—by making the acyl chloride, say, or a particularly reactive ester, which we will represent as RCOX .



The main problem, though, is that there is another free CO_2H , which could react with the COX group to form an anhydride, and two different free amines, either of which might react, giving both LeuLeu (which we don't want) and LeuGly (which we do).

For this reason, we need to protect both the NH_2 group of leucine and the CO_2H group of glycine. What sort of protecting groups do they need to be? We will need to be able to take them off again once they have done their job, so there is no point using, say, an amide to protect the amine since we would have great difficulty hydrolysing the amide in the presence of the amide bond we are trying to form. Not only do we want the protecting groups to be removable under mild conditions, but we want two groups (one for each of NH_2 and CO_2H) which we can take off under *different* conditions. We then have the opportunity to modify either end of the dipeptide at will.

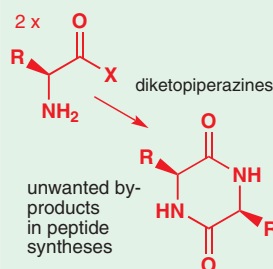


A good choice for a pair of conditions might be acid and base—we might protect the NH_2 group with a protecting group we can remove only in base, and the CO_2H group with protection we can remove only in acid.

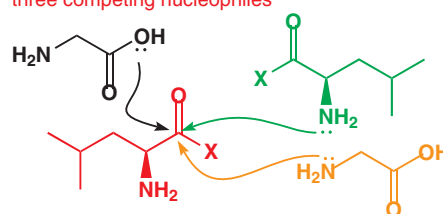


■ There is a full list of the names, structures, and abbreviation of the amino acids which constitute most peptides and proteins on p. 554.

■ In fact, acyl chlorides are not used in peptide synthesis because of the potential for unwanted side reactions, including racemization and dimerization to form by-products known as diketopiperazines.



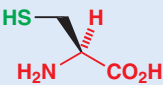
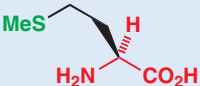
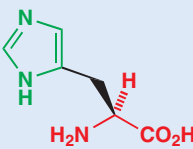
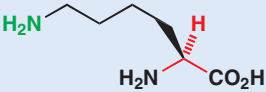
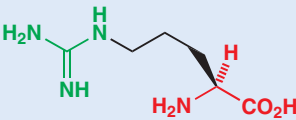
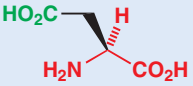
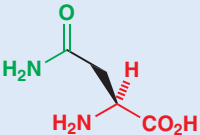
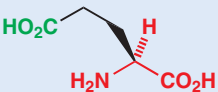
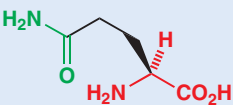
unprotected coupling—three competing nucleophiles



The amino acids

For reference, a full list of the amino acids appearing in the structure of peptides is given here, along with the codes used to describe them in abbreviated structures. The side chains are shown in black, with side chain functional groups in green. More chemistry of amino acids will follow in Chapter 42.

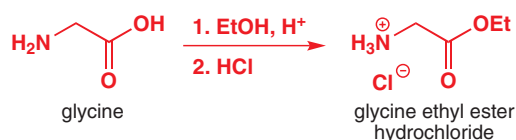
Name	Three-letter code	One-letter code	Structure
glycine	Gly	G	
alanine	Ala	A	
valine	Val	V	
leucine	Leu	L	
isoleucine	Ile	I	
phenylalanine	Phe	F	
tryptophan	Trp	W	
proline	Pro	P	
serine	Ser	S	
threonine	Thr	T	
tyrosine	Tyr	Y	

Name	Three-letter code	One-letter code	Structure
cysteine	Cys	C	
methionine	Met	M	
histidine	His	H	
lysine	Lys	K	
arginine	Arg	R	
aspartic acid	Asp	D	
asparagine	Asn	N	
glutamic acid	Glu	E	
glutamine	Gln	Q	

The Cbz protecting group—oxytocin



We introduced the dipeptide LeuGly as an example because it appears at one end of the peptide hormone oxytocin. The first step in the synthesis of oxytocin is indeed the coupling of glycine (through its amino group) with leucine. This is how it was done by du Vigneaud and Bodanszky. First, the carboxylic acid of the glycine was protected as an ethyl ester. Making an ester is the obvious way to stop CO_2H groups interfering as acids or as nucleophiles. However, simple methyl and ethyl esters may pose problems—they can still react with such nucleophiles as amines. Ethyl esters of amino acids are therefore stable only if the NH_2 group is protected. The glycine ethyl ester had to be stored as its hydrochloride salt: in effect, the -NH_2 group is ‘protected’ as -NH_3^+ .

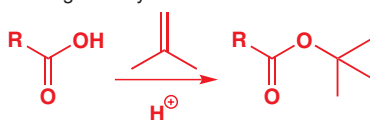


Oxytocin is a hormone involved in controlling the onset of labour in women and the subsequent release of milk. It was the first peptide hormone to be synthesized, in 1953, by du Vigneaud, who won the Nobel prize in 1955 for his work on peptide synthesis. The ‘synthetic’ version of the hormone, syntocinon (identical, of course, with the natural version isolated from human placentas, although without the dangers of biological contamination), is regularly used in modern obstetrics to induce labour in women whose babies are overdue.

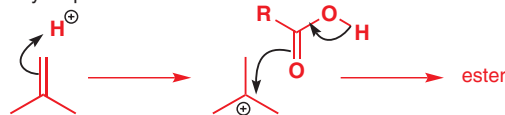
A more commonly used carboxylic-acid-protecting group that is rather more stable towards attack by nucleophiles is **the *t*-butyl ester**. *t*-Butyl esters can be made by reacting the carboxylic acid with the cation generated from isobutene in sulfuric acid.

■ This method is preferable in this case to the usual way of making esters, from acyl chloride plus alcohol (see Chapter 10), because steric hindrance makes that a very slow reaction with *t*-butanol.

making a *t*-butyl ester

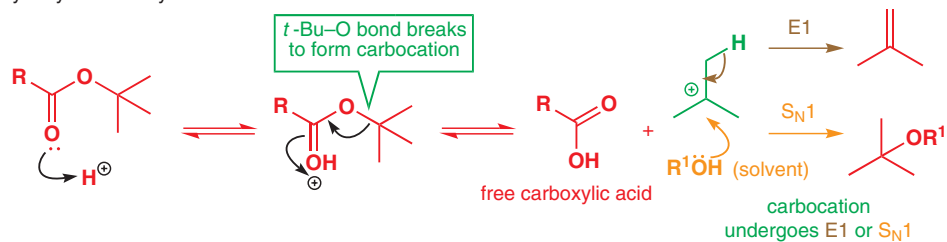


key steps:



Steric bulk means that *t*-butyl esters are resistant to nucleophilic attack at the carbonyl group, and that includes hydrolysis under basic conditions (nucleophilic attack by HO^-). But they do hydrolyse relatively easily in acid because the mechanism of hydrolysis of *t*-butyl esters in acid is quite different. Instead of undergoing nucleophilic attack at the carbonyl group, the *t*-butyl ester loses a stable carbocation, which is either captured by solvent in an $\text{S}_{\text{N}}1$ reaction or loses a proton in an $\text{E}1$ reaction.

hydrolysis of *t*-butyl esters in acid:

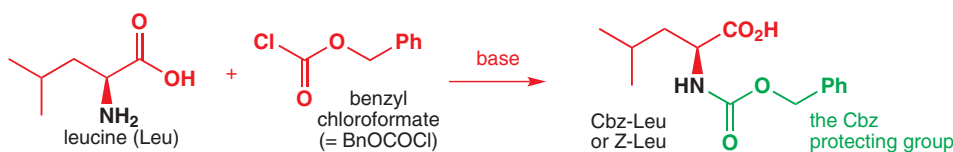


Protecting group	Structure	Protects	From	To protect	To deprotect
<i>t</i> -butyl ester ($\text{CO}_2t\text{-Bu}$)		carboxylic acids	nucleophiles	isobutene, H^+	strong acid

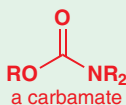
In the event, the chemists needed a group that they could later react with ammonia to make the primary amide that is present in oxytocin. They also wanted a group that was stable to mild acid—so they chose the ethyl ester.

■ You will occasionally see Cbz abbreviated to just Z.

As for the leucine residue, it had to have its NH_2 group protected using a base-stable protecting group because base would be needed to release the NH_2 group of the glycine hydrochloride salt. The group that was used is one of the most important nitrogen-protecting groups and is known as the Cbz group (Cbz stands for carboxybenzyl). Cbz groups are put on by treating with benzyl chloroformate (BnOCOCl) and weak base.

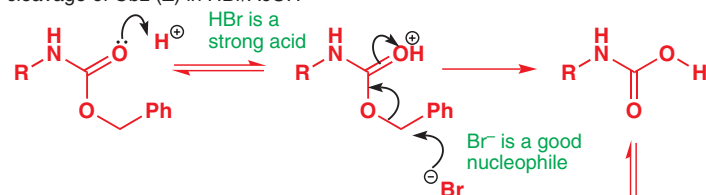


■ Carbamates are a sort of hybrid of an ester and an amide, but their chemistry resembles most closely that of an amide.

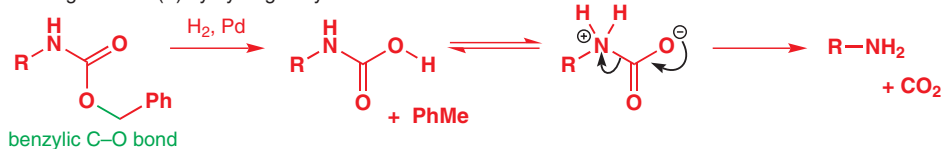


Cbz-protected amines are actually carbamates: just like amides they are no longer nucleophilic because the nitrogen's lone pair is tied up in conjugation with the carbonyl group. They are resistant to both aqueous acid and aqueous base, but they have, to use the analogy we developed earlier, an Achilles' heel—the benzyl group. Removal of the benzyl group under the same two sets of conditions that remove benzyl ethers (p. 551) releases the safety catch and removes Cbz:

cleavage of Cbz (Z) in HBr/AcOH

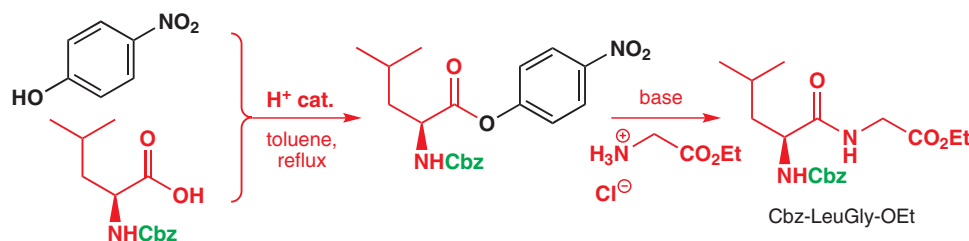


cleavage of Cbz (Z) by hydrogenolysis



Protecting group	Structure	Protects	From	Protection	Deprotection
Cbz (Z) (OCOBn)		amines	electrophiles	BnOCOCl, base	HBr, AcOH; or H ₂ , Pd

The carboxylic acid of the Cbz-protected leucine next has to be activated to allow it to react with the glycine. The acyl chloride won't do as it is unstable, and an alternative in peptide chemistry is to make a *p*-nitrophenyl or 2,4,6-trichlorophenyl ester. Phenoxide, especially when substituted with electron-withdrawing substituents, is a good leaving group, and Cbz-leucine *p*-nitrophenyl ester reacts with the glycine hydrochloride ethyl ester in the presence of a weak base (triethylamine, to release the glycine's NH₂ group).



■ The lower pK_a of a phenol makes a phenoxide a better leaving group than an alkoxide, and an electron-deficient phenoxide is even better still (see p. 173).

■ Notice the chemoselectivity in this step—the glycine's NH₂ group has three carbonyl groups to choose from, but reacts only with the most electrophilic—the one bearing the best leaving group.

The dipeptide is now coupled—but is still protected. Deprotection (HBr/AcOH) gave the HCl salt of LeuGly ethyl ester for further reaction. The rest of the peptide was built up in much the same way—each amino acid being introduced as the Cbz-protected *p*-nitrophenyl ester before being deprotected ready for the next coupling, until all nine of oxytocin's amino acids had been introduced.

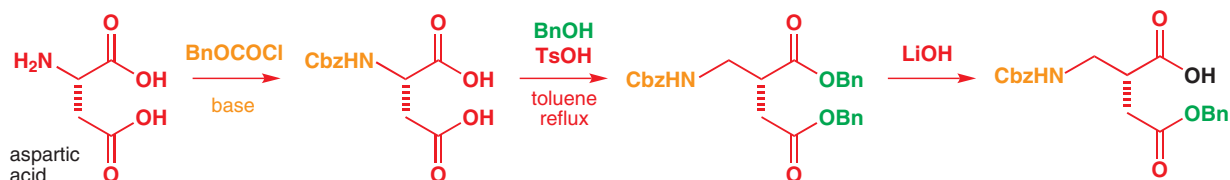
The Boc protecting group—gastrin and aspartame

Gastrin is a hormone released from the stomach that controls the progress of digestion. Early work on the hormone showed that only the four C-terminal amino acids of the peptide (the C-terminal tetrapeptide) were necessary for its physiological activity.

The synthesis starts with the coupling of two more amino acids: aspartic acid and phenylalanine. As you would expect, the carboxylic acid group of phenylalanine is protected, this time as a methyl ester, and the NH₂ group of aspartic acid is protected as a Cbz derivative. Since aspartic acid has two carboxylic acid groups, one of these also has to be protected. Here is the method—first the Cbz group is put on; then both acids are protected as *benzyl* esters. Then just one of the benzyl esters is hydrolysed. It may seem surprising to you that this chemoselective hydrolysis is possible, and you could not have predicted that it would work without trying it out in the laboratory.

H₂N–Tyr–Met–Asp–Phe–CONH₂
gastrin C-terminal tetrapeptide

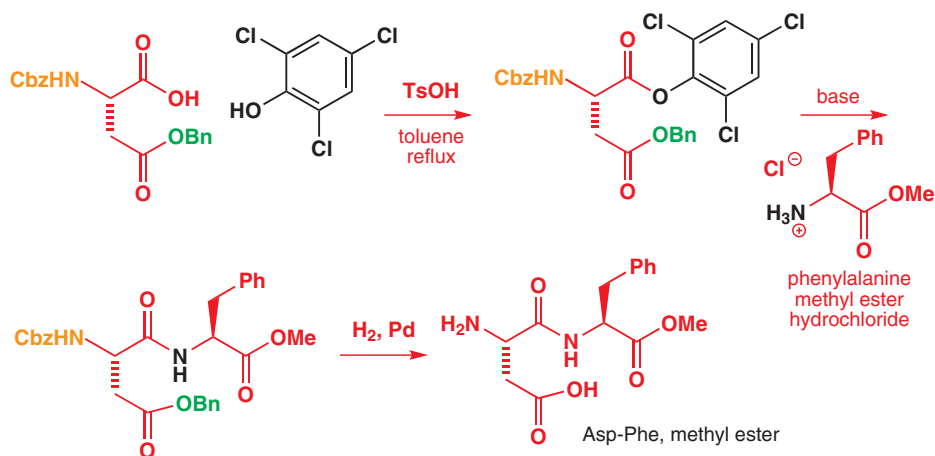
C-terminus means the end of the peptide that carries the terminal CO₂H group. The other end, carrying the NH₂ group, is the N-terminus. By convention, the N-terminus is always written to the left and the C-terminus to the right.



Accidental aspartame

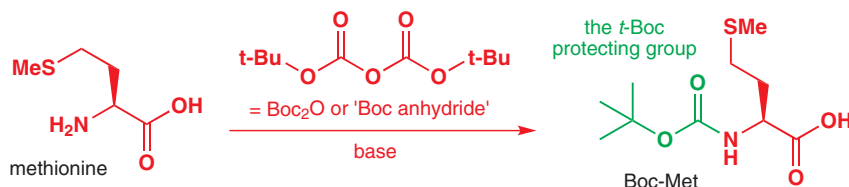
At this point in one synthesis of the tetrapeptide in the laboratories of Searle, the now defunct American pharmaceutical company, a remarkable discovery occurred. The AspPhe methyl ester was accidentally found to taste sweet: extremely sweet—about 200 times as sweet as sucrose. AspPhe is now known as aspartame, marketed under the brand name NutraSweet. It goes without saying that despite this extraordinary discovery, tasting anything in the laboratory, accidentally or otherwise, is extremely unwise, ill-judged, and outright dangerous. Donald Rumsfeld was once chief executive of Searle.

The protected acid is next activated as its 2,4,6-trichlorophenyl ester, ready for coupling with the phenylalanine methyl ester in base. Now you see why the benzyl ester was chosen to protect Asp's side-chain carboxylic acid group—hydrogenolysis can be used to cleave both the Cbz group and the benzyl ester at the same time.



The next amino acid in the peptide is methionine, and it will of course need N-protecting and C-activating. The N-protecting group used this time was different—still a carbamate, although not Cbz—it was Boc, which stands for *t*-butoxycarbonyl and is pronounced 'bock'. The Boc group, *t*-BuOCO, is introduced with (*t*-BuOCO)₂O, known as Boc anhydride.

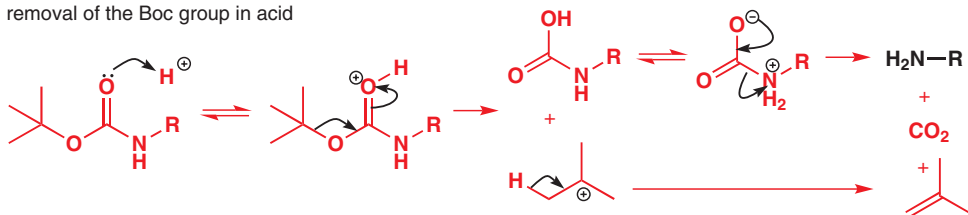
■ To introduce the Cbz group, a chloroformate (BnOCCl) is used. But the equivalent chloroformate here, $t\text{-BuOCCl}$, is unstable, so Boc_2O is used instead. It would be good practice for you to write a mechanism for this reaction.



Like Cbz, the Boc group is a carbamate protecting group. But, unlike Cbz, it can be removed simply with dilute aqueous acid. Just 3M HCl will hydrolyse it, again by protonation, loss of *t*-butyl cation, and decarboxylation. Base, on the other hand, cannot touch the Boc group—the carbonyl group is too hindered to be attacked even by OH^- , and Boc is strongly resistant to basic hydrolysis.

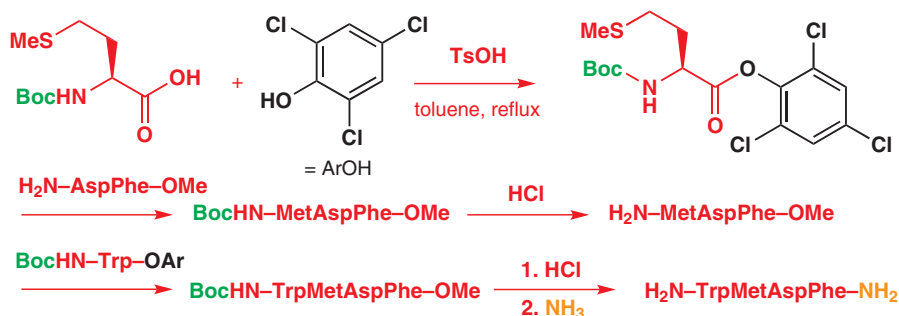
■ The mechanism for this hydrolysis is comparable to the acid-catalysed cleavage of Cbz groups, but remember that here the *t*-Bu group leaves in an $\text{S}_{\text{N}}1$ step. Cbz groups are cleaved by using a good nucleophile, Br^- , because an $\text{S}_{\text{N}}2$ step is involved; any old acid will cleave Boc.

removal of the Boc group in acid



Protecting group	Structure	Protects	From	To protect	To deprotect
Boc (<i>t</i> -BuOCO)		amines	electrophiles	(<i>t</i> -BuOCO) ₂ O, base	H ⁺ , H ₂ O

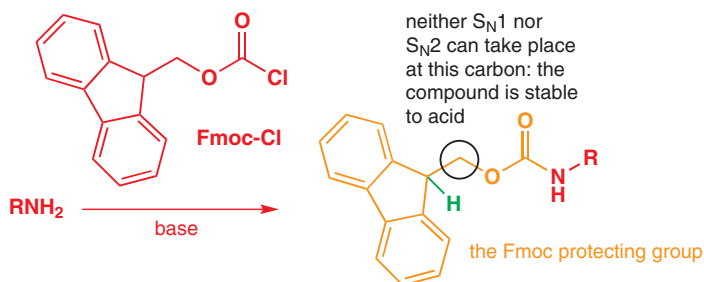
Meanwhile, back at the tetrapeptide synthesis, methionine (Met) has been Boc-protected, and is ready for activation—as a 2,4,6-trichlorophenyl ester (abbreviated to Ar below) this time—and coupling with the deprotected AspPhe-OMe. Aqueous acid takes off the Boc group without hydrolysing peptide or ester bonds, and a repeat of this cycle with Boc-tryptophan trichlorophenyl ester (BocHN-Trp-OAr) followed by formation of the amide with ammonia finally gives the tetrapeptide.



The Fmoc protecting group

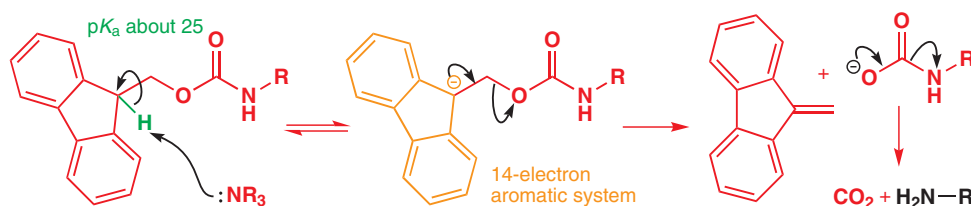
Our final protecting group has a susceptibility inverse to that of Boc. The Fmoc (pronounced 'eff-mock'), or fluorenylmethyloxycarbonyl, protecting group cannot be lost by substitution in the manner of Cbz or *t*-Boc because neither S_N1 nor S_N2 mechanisms can operate at the ringed carbon atom: it is both primary *and* hindered.

➡ In Chapter 15 we analysed in detail the structural features which favour and disfavour substitution reactions of this type.



So, where is the safety catch? Fmoc's Achilles' heel is its rather acidic proton (pK_a about 25), shown in green. It's acidic because the anionic product of deprotonation is aromatic. Only a very small concentration of this aromatic anion ever forms, but as it does it immediately undergoes elimination. Fmoc-protected amines are readily deprotected in base.

➡ The aromaticity of cyclopentadienyl anions of this type was discussed in Chapter 17, p. 401.



The table of protecting groups, built up slowly over this chapter, is now complete. You should from this point on be able to write a structure for each of the ones listed below, and you should also be familiar with the types of conditions necessary for protection and deprotection with each member of the list.

Protecting group	Structure	Protects	From	To protect	To deprotect
acetal (dioxolane))		ketones, aldehydes	nucleophiles, bases		H ⁺ , H ₂ O
trialkylsilyl R ₃ Si (e.g. TBDMS)	RO-SiMe ₃ RO-SiMe ₂ t-Bu	alcohols (OH in general)	nucleophiles, C or N bases	R ₃ SiCl, base	H ⁺ , H ₂ O, or F ⁻
tetrahydropyranyl (THP)		alcohols (OH in general)	strong bases		H ⁺ , H ₂ O
benzyl ether (OBn)		alcohols (OH in general)	almost everything	NaH, BnBr	H ₂ , Pd/C, or HBr
methyl ether (ArOMe)		phenols (ArOH)	bases	NaH, MeI, or (MeO) ₂ SO ₂	BBr ₃ , HBr, HI, Me ₃ SiI
t-butyl ester (CO ₂ t-Bu)		carboxylic acids	nucleophiles	isobutene, H ⁺	strong acid
Cbz (Z) (OCOBn)		amines	electrophiles	BnOCOCl, base	HBr, AcOH; or H ₂ , Pd
t-Boc (OCOt-Bu)		amines	electrophiles	(t-BuOCO) ₂ O, base	H ⁺ , H ₂ O
Fmoc	see text	amines	electrophiles	Fmoc-Cl	base, e.g. amine

Chemoselective methods for oxidation and reduction, and protecting groups to help control chemoselectivity, will appear throughout this book, and we shall return in detail to peptides and their biological functions in Chapter 42. Before then we will address in detail stereoselectivity (in Chapters 32, 33, and 41) but the very next chapter will deal with the other aspect of selectivity—*regioselectivity*.

Further reading

There is a basic introduction in S. Warren and P. Wyatt, *Organic Synthesis: the Disconnection Approach*, Wiley, Chichester, 2008, chapter 5.

Molecular Orbitals and Organic Chemical Reactions: Student Edition by Ian Fleming, Wiley, Chichester, 2009 on pentadienyl anions, pp. 126–128.

Birch reduction: P. W. Rabideau, and Z. Marcinow, *Org. React.* 1992, **42**, 1. Lindlar reduction: H. Lindlar and R. Dubuis, *Org. Synth. Coll.* 1973, vol 5, 880.

Peptide synthesis: N. L. Benoiton, *Chemistry of Peptide Synthesis*, Taylor and Francis, 2005. J. Jones, *Amino Acid and Peptide Synthesis*, Oxford Primer, 2nd edn, OUP, Oxford, 2002.

Protecting Groups: basic introduction, Jeremy Robertson, *Protecting Group Chemistry*, Oxford Primer, Oxford, OUP, 2000. More advanced books: P. J. Kocienski, *Protecting Groups*, 3rd edn, Thieme, 2003. P. G. M. Wuts and T. Greene, *Greene's Protecting Groups in Organic Synthesis*, Wiley, 2007. A different view: T. Newhouse, P. S. Baran, and R. W. Hoffmann, The Economies of Synthesis, *Chem. Soc. Rev.* 2009, **38**, 3010.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>